COSC343/AIML402

MODULE: MACHINE LEARNING

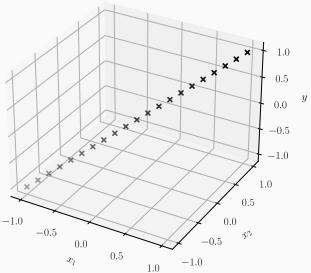
LECTURE 14b: OPTIMISATION

Lech Szymanski and Veronica Liesaputra

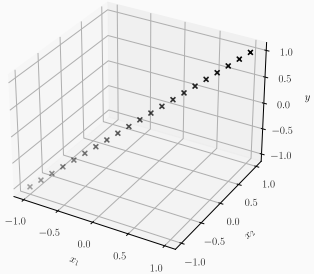
In this lecture:

- **Learning as state space search**
- **Linear models**
- **Parameter search**

Starting problem: Perfect regression?!?



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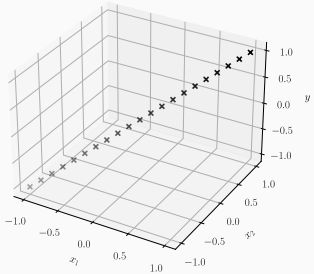


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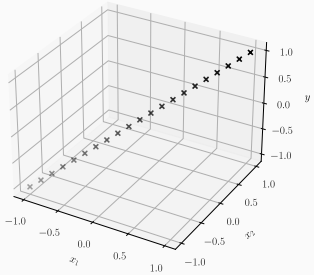
Let's use Least Squares to fit data with the model:

$$\hat{y} = w_1 x_1 + w_2 x_2 + w_0$$



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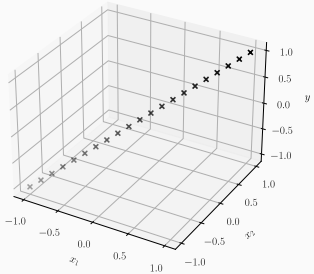
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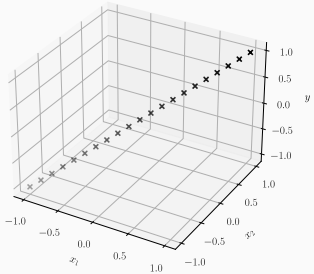
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you get a Singular matrix exception!!!

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- $Det(M) = 0$.
- Thus, M is an example of singular matrix.

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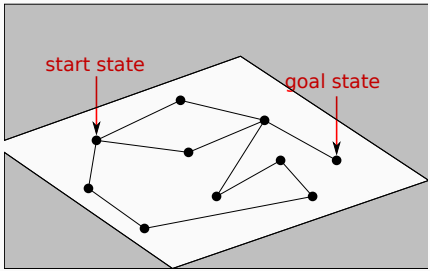
$$M^{-1} = \frac{Adj(M)}{Det(M)}$$
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We cannot divide by 0, so M^{-1} is infinite! or we can say that M is not invertible.

Review: State space search

In a **state-space search** task,

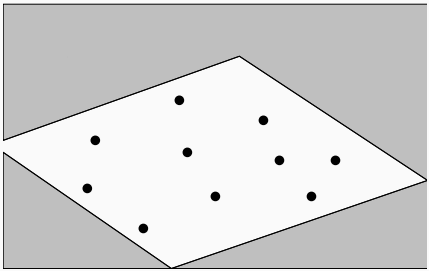
- There's a set of possible states;
- To get into a given state, you may have to go through other states.
- You know which state is best: but you don't know how to get there...



Review: State space search

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- There's a set of possible states;
- ~~To get into a given state, you may have to go through other states.~~
- ~~You know which state is best: but you don't know how to get there...~~
- ...though sometimes all we want is to find the best possible state according to some criteria.



Learning as state space search

Learning as state space search

The learning approach to AI, given a dataset made of samples (x, y), can be characterised as follows:

- we make a hypothesis $\hat{y} = h(x, w)$,
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- The states space is (usually) continuous.
- The loss function evaluates each state.

Loss surface

When we're working with a specific learning problem, the training dataset is fixed – we have a set of points (x, y) to work with, but that set is constant.

Given that constant dataset, we can think of the loss J evaluated on $h(x, w)$ for some choice of parameters w as a function $J(w)$ in the space of all possible parameters (the dimensionality of the parameter space is equal to the number of parameters in w).

The manifold that $J(w)$ makes (for a given dataset) in the parameter space is referred to as the **loss surface** (also **loss function** and **loss landscape**)

Visualisation of the loss surface

Recall the regression example from the previous lecture: $h(x, w) = w_1 x + w_0$

$\hat{y} = 3.4x - 3.4$, $\hat{y} = 1.7x + 4.6$, $\hat{y} = 1.7x + 6.7$

x	y	$\hat{y}$	$y - \hat{y}$	$\hat{y}$	$y - \hat{y}$	$\hat{y}$	$y - \hat{y}$
1.19	7.16	0.67	6.49	6.63	0.53	8.63	-1.47
1.63	5.87	2.16	3.71	7.38	-1.51	9.38	-3.51
2.24	7.12	4.26	2.86	8.42	-1.30	10.42	-3.30
2.55	10.94	5.33	5.61	8.96	1.98	10.96	-0.02
2.76	13.22	6.05	7.18	9.32	3.90	11.32	1.90
3.4	6.96	8.25	-1.29	10.42	-3.46	12.42	-5.46
4.98	11.44	13.66	-2.23	13.13	-1.69	15.13	-3.69
5.06	13.70	13.92	-0.22	13.26	0.45	15.26	-1.55
5.47	15.34	15.34	0.00	13.96	1.37	15.96	-0.63
5.85	15.68	16.64	-0.96	14.62	1.07	16.62	-0.93
6.46	12.94	18.73	-5.79	15.66	-2.72	17.66	-4.72
6.55	17.60	19.03	-1.43	15.81	1.79	17.81	-0.21
6.8	15.36	19.88	-4.51	16.23	-0.87	18.23	-2.87
7	17.38	20.54	-3.16	16.56	0.81	18.56	-1.19
7.09	15.70	20.89	-5.19	16.74	-1.04	18.74	-3.04
7.51	17.63	22.33	-4.69	17.46	0.17	19.46	-1.83
7.55	15.97	22.44	-6.47	17.52	-1.55	19.52	-3.55
8.91	23.99	27.11	-3.12	19.85	4.14	21.85	2.14
9.59	20.58	29.45	-8.88	21.02	-0.45	23.02	-2.45
9.6	19.39	29.47	-10.08	21.03	-1.64	23.03	-3.64
J_{MSE}							

$$J_{\text{MSE}}(w_1, w_0) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

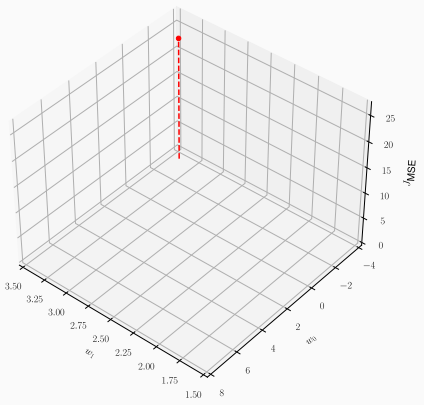
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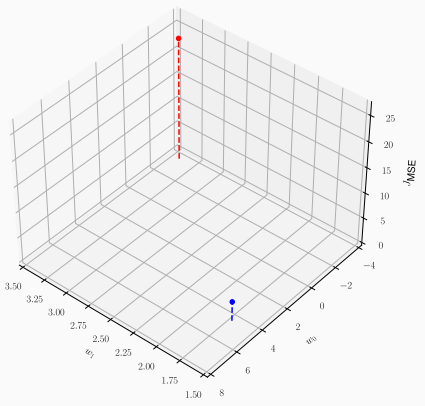
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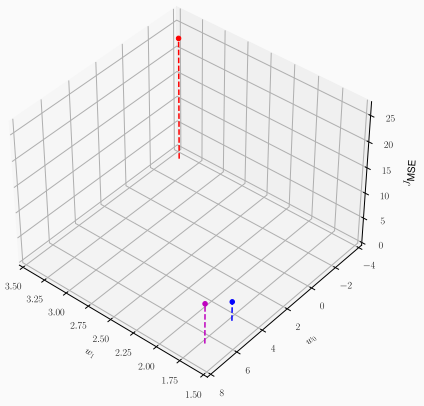
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5.47	15.34	15.34	0.00	13.96	1.37	15.96	-0.63
5.85	15.68	16.64	-0.96	14.62	1.07	16.62	-0.93
6.46	12.94	18.73	-5.79	15.66	-2.72	17.66	-4.72
6.55	17.60	19.03	-1.43	15.81	1.79	17.81	-0.21
6.8	15.36	19.88	-4.51	16.23	-0.87	18.23	-2.87
7	17.38	20.54	-3.16	16.56	0.81	18.56	-1.19
7.09	15.70	20.89	-5.19	16.74	-1.04	18.74	-3.04
7.51	17.63	22.33	-4.69	17.46	0.17	19.46	-1.83
7.55	15.97	22.44	-6.47	17.52	-1.55	19.52	-3.55
8.91	23.99	27.11	-3.12	19.85	4.14	21.85	2.14
9.59	20.58	29.45	-8.88	21.02	-0.45	23.02	-2.45
9.6	19.39	29.47	-10.08	21.03	-1.64	23.03	-3.64
J_{MSE}			25.07		3.85		7.85

$$J_{\text{MSE}}(w_1, w_0) = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$



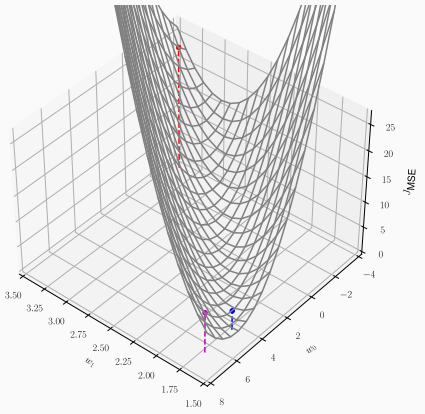
Visualisation of the loss surface

Recall the regression example from the previous lecture: $h(x, w) = w_1 x + w_0$

$\hat{y} = 3.4x - 3.4$, $\hat{y} = 1.7x + 4.6$, $\hat{y} = 1.7x + 6.7$

x	y	$\hat{y}$	$y - \hat{y}$	$\hat{y}$	$y - \hat{y}$	$\hat{y}$	$y - \hat{y}$
1.19	7.16	0.67	6.49	6.63	0.53	8.63	-1.47
1.63	5.87	2.16	3.71	7.38	-1.51	9.38	-3.51
2.24	7.12	4.26	2.86	8.42	-1.30	10.42	-3.30
2.55	10.94	5.33	5.61	8.96	1.98	10.96	-0.02
2.76	13.22	6.05	7.18	9.32	3.90	11.32	1.90
3.4	6.96	8.25	-1.29	10.42	-3.46	12.42	-5.46
4.98	11.44	13.66	-2.23	13.13	-1.69	15.13	-3.69
5.06	13.70	13.92	-0.22	13.26	0.45	15.26	-1.55
5.47	15.34	15.34	0.00	13.96	1.37	15.96	-0.63
5.85	15.68	16.64	-0.96	14.62	1.07	16.62	-0.93
6.46	12.94	18.73	-5.79	15.66	-2.72	17.66	-4.72
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7.55	15.97	22.44	-6.47	17.52	-1.55	19.52	-3.55
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J_{MSE}			25.07		3.85		7.85

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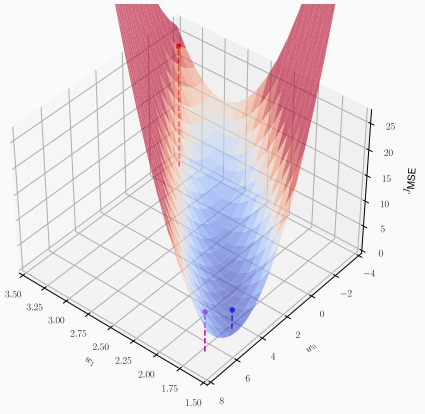
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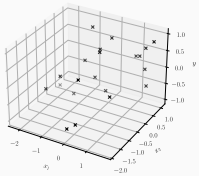
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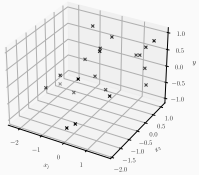


The problem with Least Squares

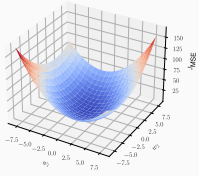


$\hat{y} = w_1 x_1 + w_2 x_2$

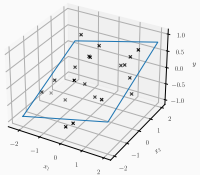
The problem with Least Squares



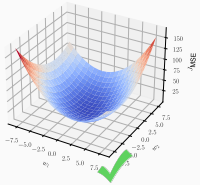
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The problem with Least Squares

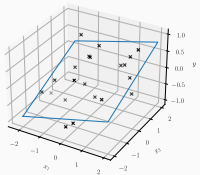


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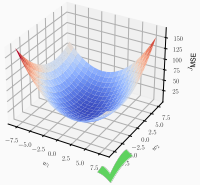


$$\mathbf{w} = (\bar{X}^T \cdot \bar{X})^{-1} \cdot \bar{X}^T \cdot \mathbf{y}$$

The problem with Least Squares

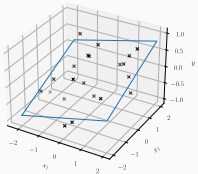


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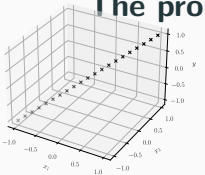


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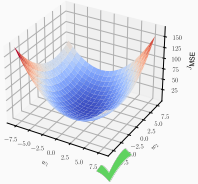
The problem with Least Squares



$\hat{y} = w_1 x_1 + w_2 x_2$

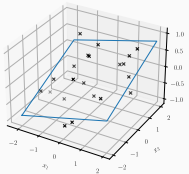


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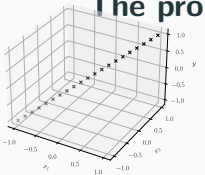


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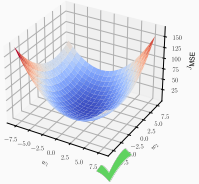
The problem with Least Squares

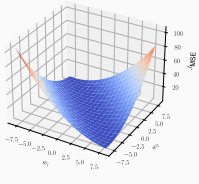


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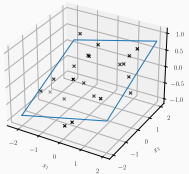
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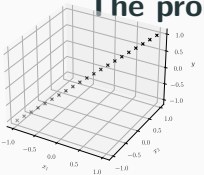


$$\mathbf{w} = (\bar{X}^T \cdot \bar{X})^{-1} \cdot \bar{X}^T \cdot \mathbf{y}$$

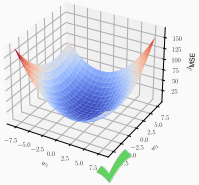
The problem with Least Squares



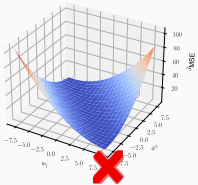
$\hat{y} = w_1 x_1 + w_2 x_2$



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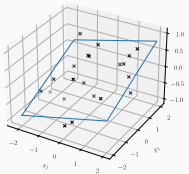


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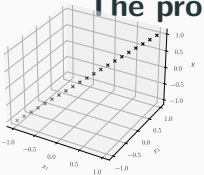
Least squares works only when:

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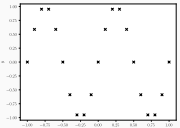
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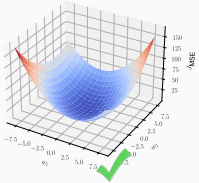
$\hat{y} = w_1 x_1 + w_2 x_2$



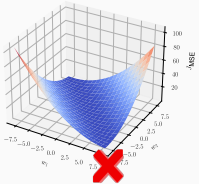
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$\hat{y} = w_1 \sin(w_2 x_2)$



$\mathbf{w} = (\bar{X}^T \cdot \bar{X})^{-1} \cdot \bar{X}^T \cdot \mathbf{y}$

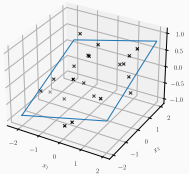


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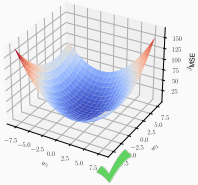
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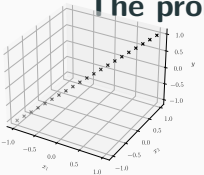
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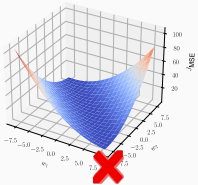
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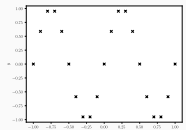
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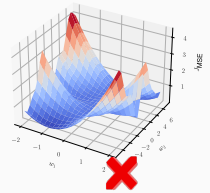
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$\mathbf{w} = (\bar{X}^T \cdot \bar{X})^{-1} \cdot \bar{X}^T \cdot \mathbf{y}$



$\hat{y} = w_1 \sin(w_2 x_2)$

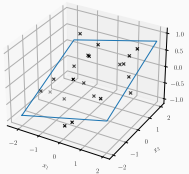


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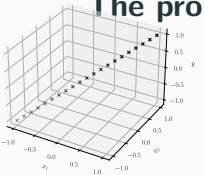
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- The model is linear in parameters

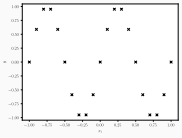
The problem with Least Squares



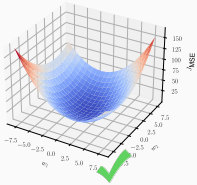
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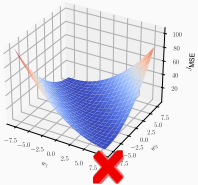
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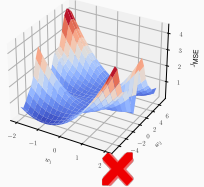
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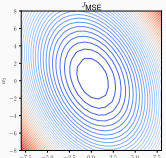
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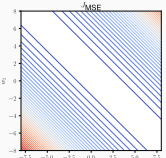


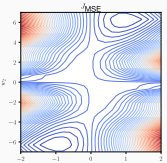
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Least squares works only when:

- $(\bar{X}^T \cdot \bar{X})$ is non-singular (invertible)
- The model is linear in parameters

Linear models

Linear model

A hypothesis $h(x, w)$ of k parameters that has the following format:

$$\hat{y} = h(x, w) = w_1 g_1(x) + w_2 g_2(x) + \ldots w_k g_k(g),$$

where $g_i(x)$ can be any function, is said to be **linear in parameters**.

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Example: Polynomial hypothesis of degree k

$$\hat{y} = h(x, w) = \sum_{i=0}^{k} w_i g_i(x),$$

where $g_i(x) = x^i$, is linear in parameters.

Non-linear model

A **non-linear** hypothesis $h(x, w)$ is some nonlinear function of w.

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Example: The following hypothesis

$$\hat{y} = h(x, w) = \sin(w_1 x_1 + w_2 x_2) + w_0$$

is non-linear in parameters.

Hypothesis linearity and the nature of the loss surface

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Hence, Least Squares is limited to only certain types of problems and linear hypotheses. We need a method for finding model parameters that is more generic.

Parameter search

Parameter search

For most of the *interesting* learning problems it's (computationally) too expensive to compute the loss for a wide range of possible values that parameters can take

- Models have hundreds, thousands, even millions of parameters – computing J over a regularly sampled range of parameters is impossible

Hence, we need some kind of search...that aims to find a state of parameters that minimise the loss, which is equivalent to finding a **minimum** point on the loss surface.

Optimisation techniques are used to search for parameters that give the minimum loss.

Formalising parameter search

Given a dataset of pairs of points (x, y), a choice of model $h(x, w)$ and a loss function $J(w)$:

1. Initialise w to random values
2. For some number of epochs
3. Evaluate the output of $h(x, w)$
4. Evaluate the loss $J(w)$
5. Compute the parameter update Δw
6. Update parameters $w := w + \alpha \Delta w$
7. The final value of w is the final state

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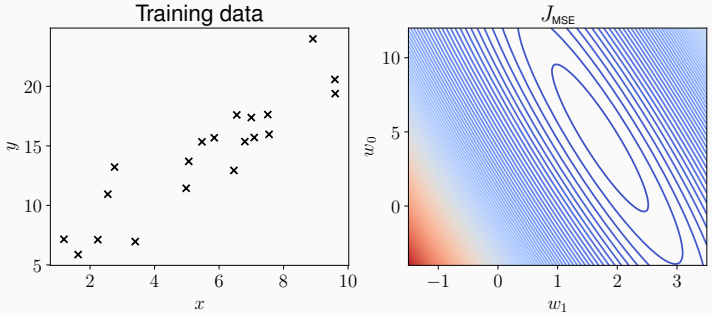
The trick is to pick Δw every time so that $J(w)$ keeps decreasing.

Random walk optimisation

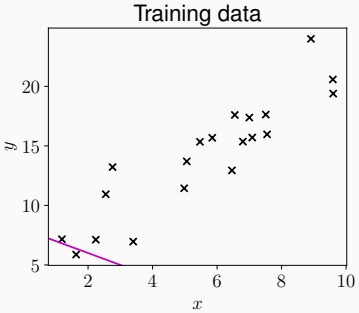
Random walk is an uninformed parameter search. You pick Δw at random, update the parameters only if:

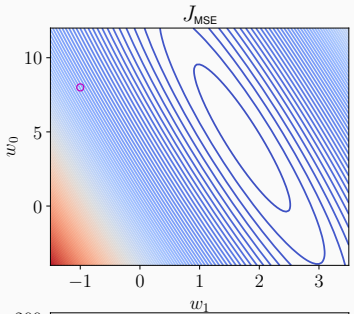
$$J(w + \alpha \Delta w) < J(w)$$

Example: Random walk



Example: Random walk





$\hat{y} = w_1 x + w_0$

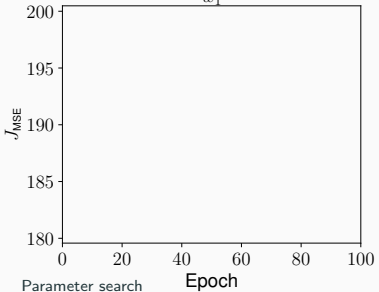
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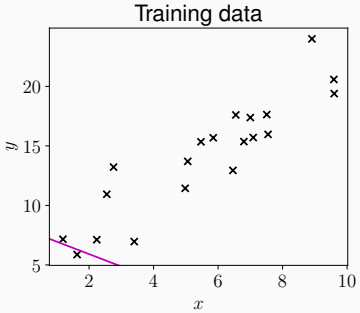
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

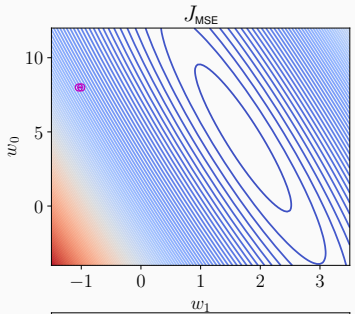
$w_1 = -1.0, w_0 = 8$

$J_{\text{MSE}} = 190.03$



Example: Random walk





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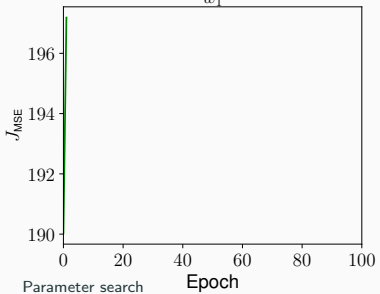
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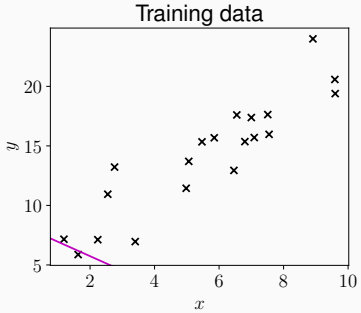
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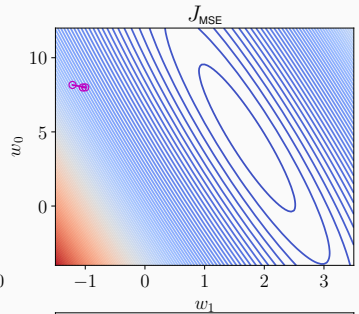
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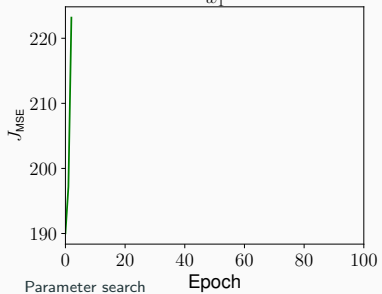
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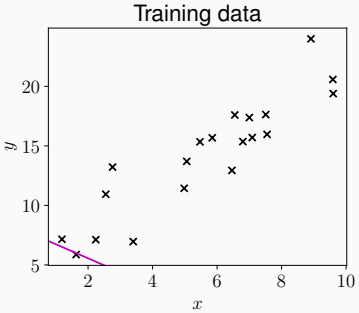
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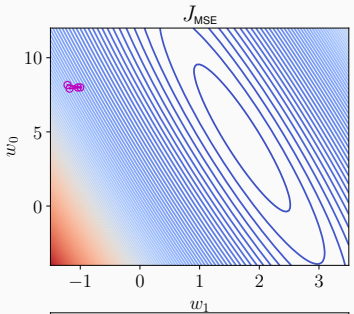
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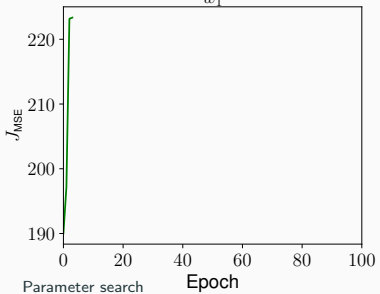
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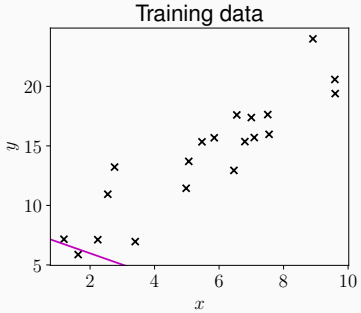
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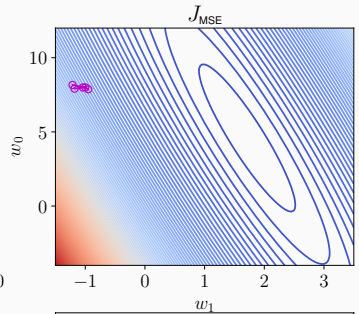
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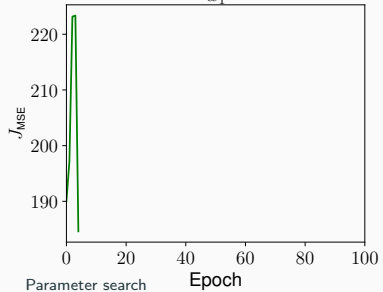
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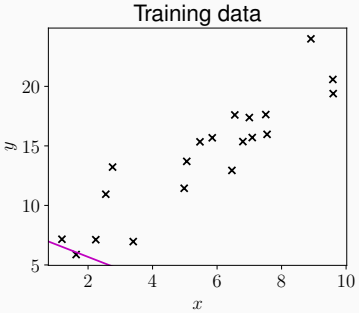
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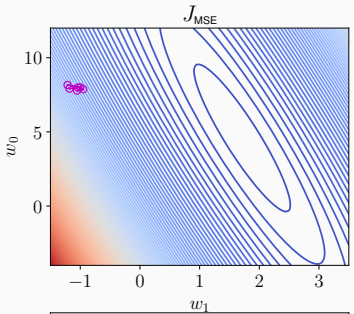
$w_1 = -0.9, w_0 = 8$

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Example: Random walk





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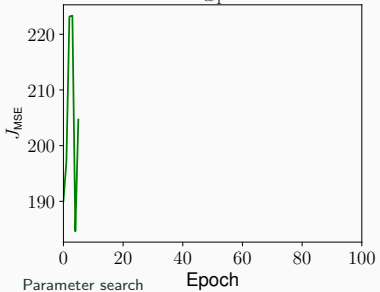
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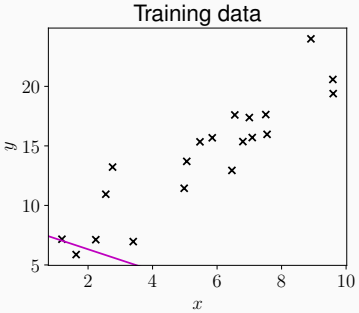
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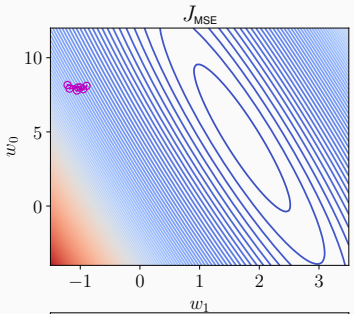
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Example: Random walk





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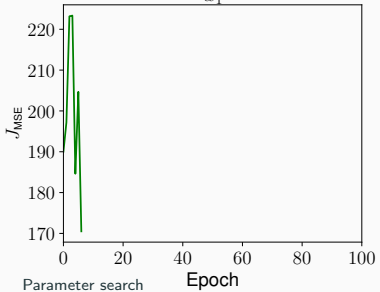
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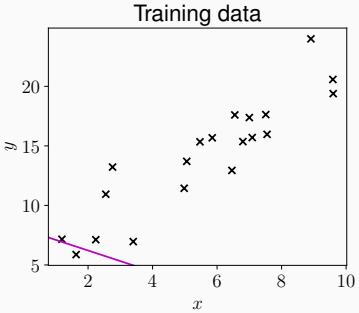
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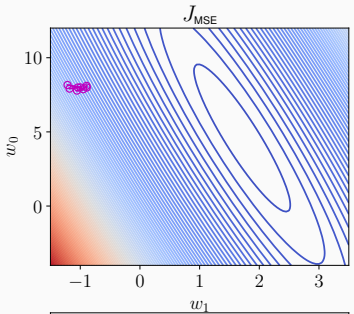
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Example: Random walk





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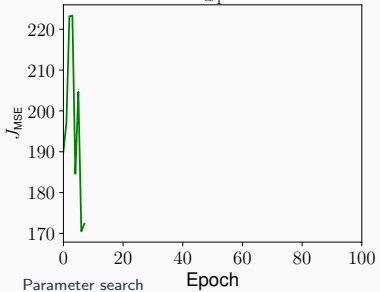
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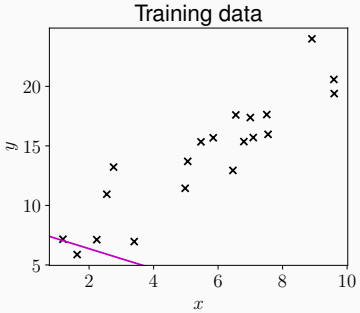
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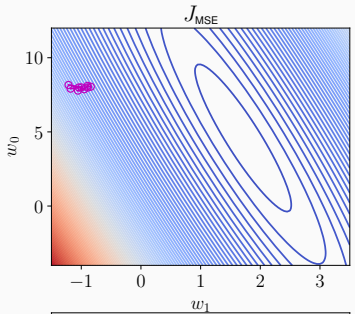
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Example: Random walk





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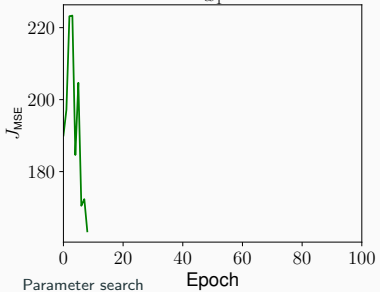
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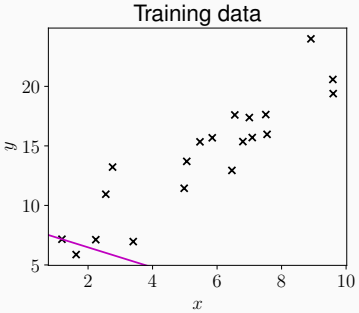
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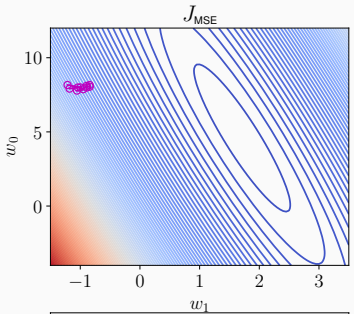
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 163.40$



Example: Random walk





$\hat{y} = w_1 x + w_0$

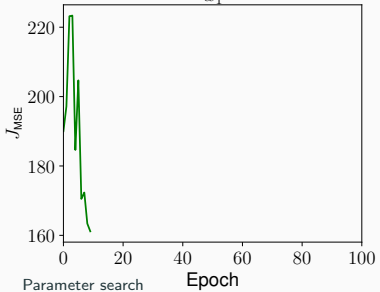
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

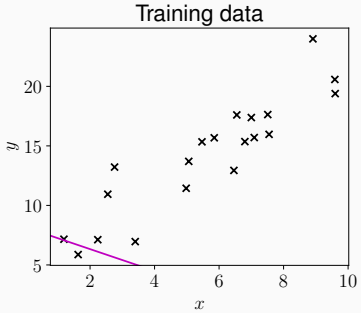
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

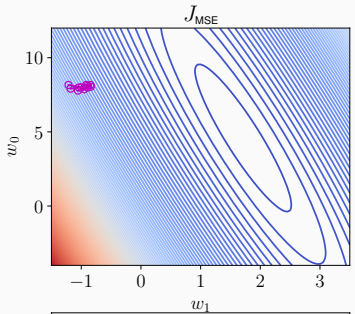
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 161.16$



Example: Random walk





$\hat{y} = w_1 x + w_0$

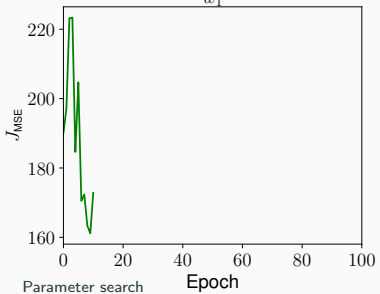
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

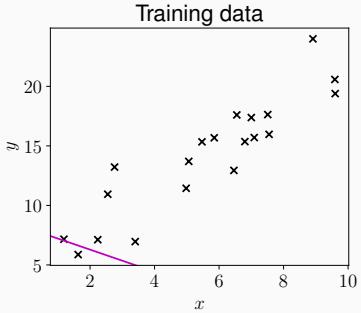
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

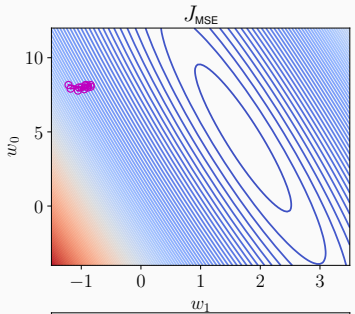
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 161.16$



Example: Random walk





$\hat{y} = w_1 x + w_0$

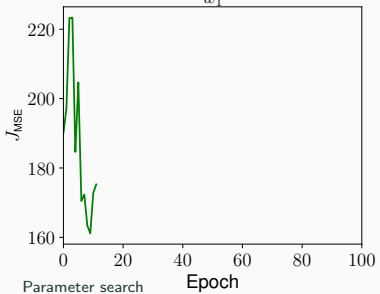
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

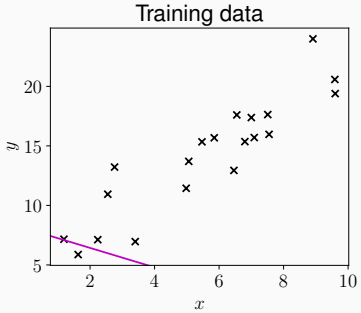
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

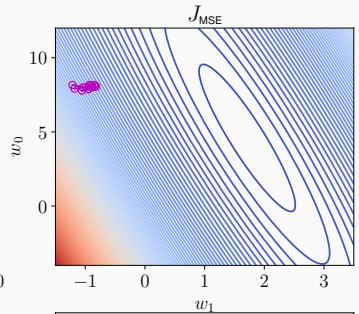
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 161.16$



Example: Random walk





$\hat{y} = w_1 x + w_0$

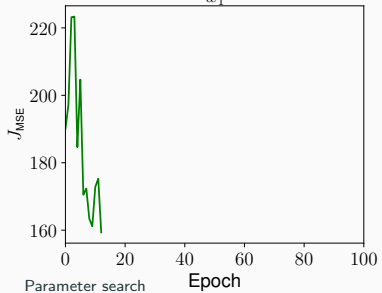
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

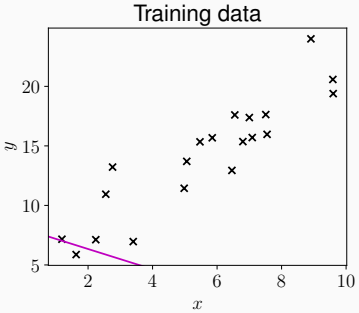
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

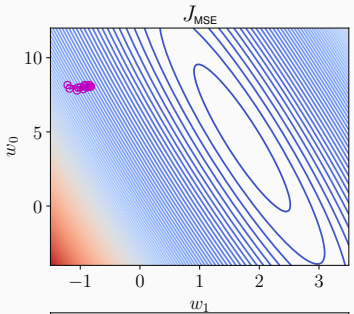
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 159.36$



Example: Random walk





$\hat{y} = w_1 x + w_0$

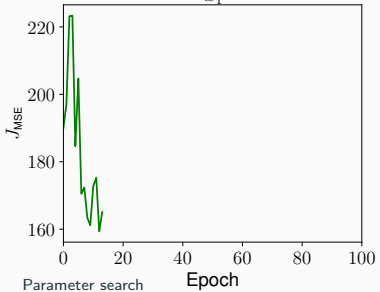
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

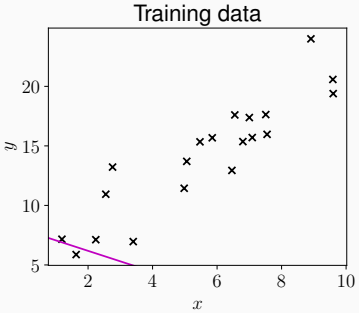
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

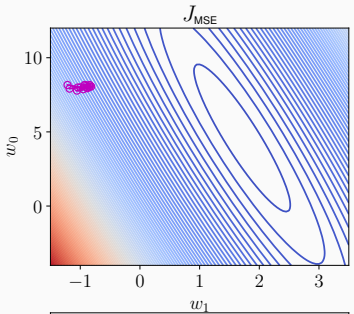
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 159.36$



Example: Random walk





$\hat{y} = w_1 x + w_0$

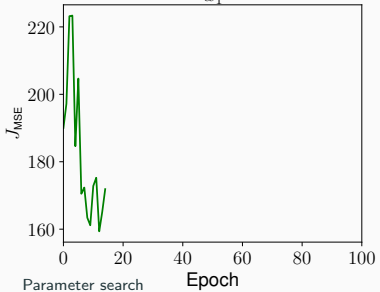
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

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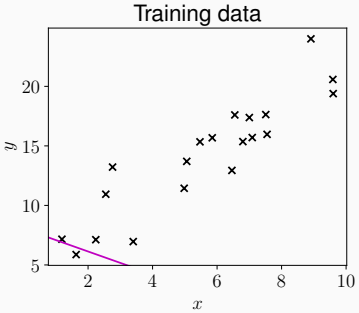
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

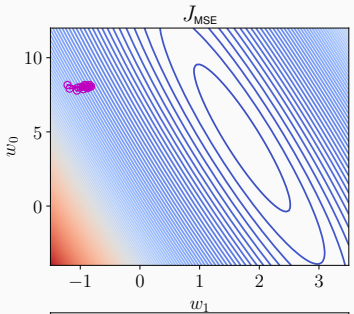
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 159.36$



Example: Random walk





$\hat{y} = w_1 x + w_0$

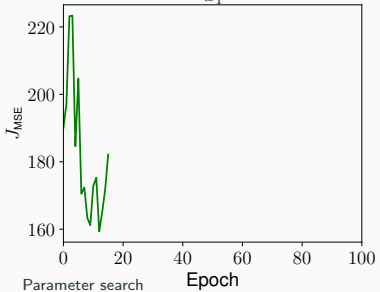
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

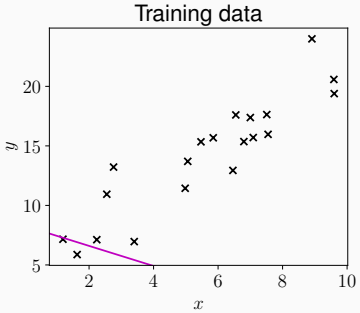
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

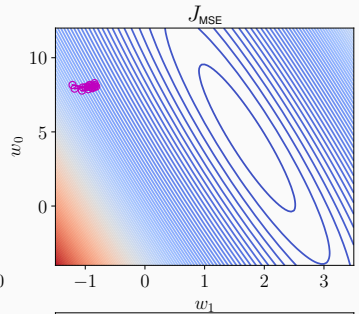
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 159.36$



Example: Random walk





$\hat{y} = w_1 x + w_0$

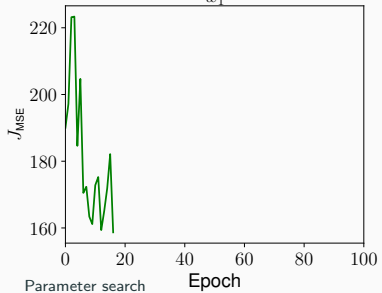
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

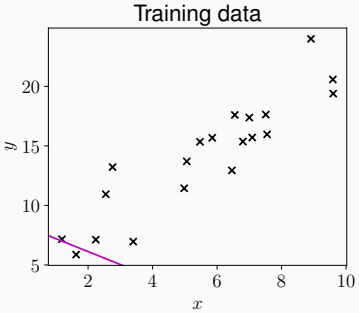
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

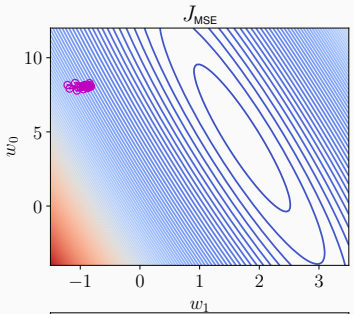
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 158.69$



Example: Random walk





$\hat{y} = w_1 x + w_0$

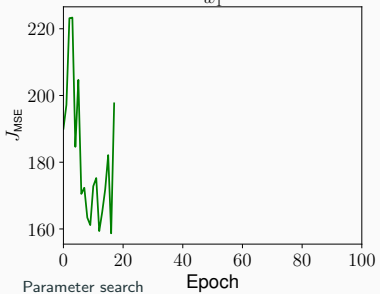
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

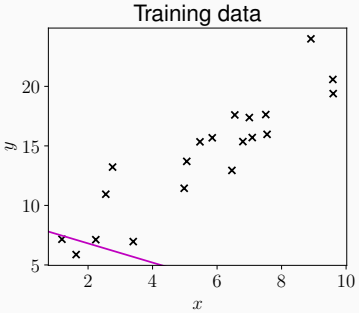
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

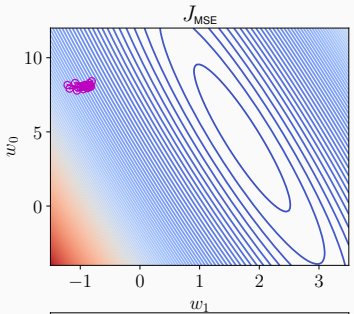
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 158.69$



Example: Random walk





$\hat{y} = w_1 x + w_0$

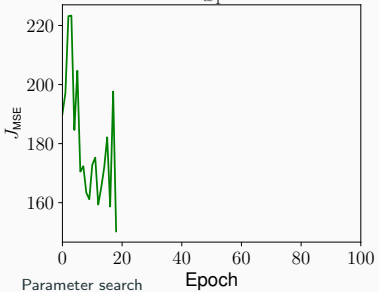
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

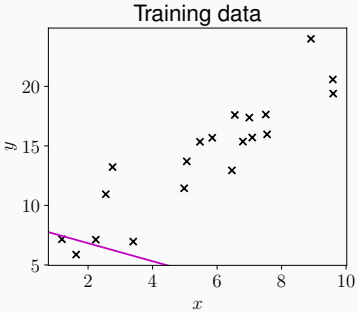
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

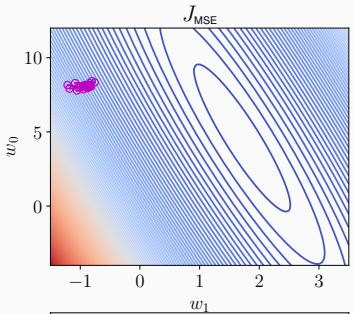
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 150.29$



Example: Random walk





$\hat{y} = w_1 x + w_0$

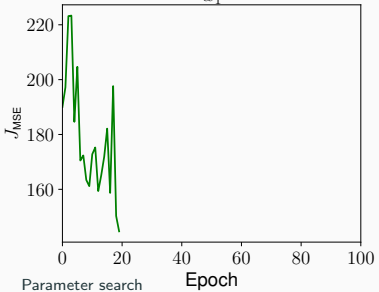
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

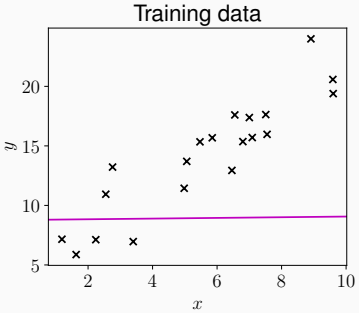
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

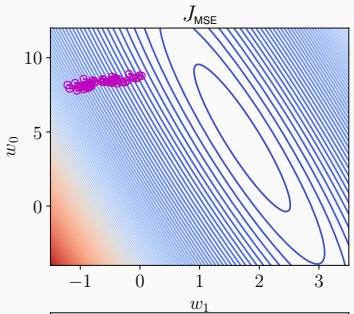
$w_1 = -0.8, w_0 = 8$

$J_{\text{MSE}} = 144.68$



Example: Random walk





$\hat{y} = w_1 x + w_0$

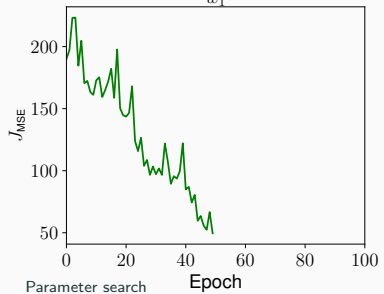
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

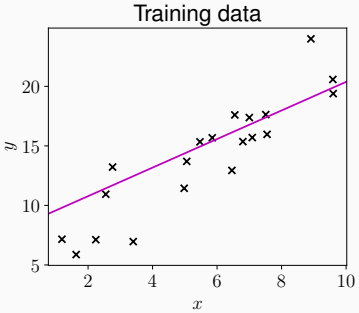
$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

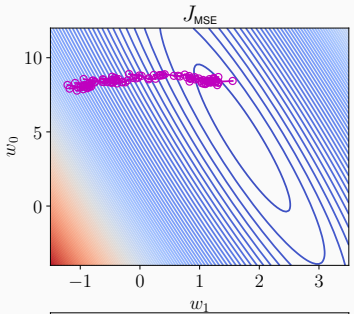
$w_1 = 0.0, w_0 = 9$

$J_{\text{MSE}} = 49.53$



Example: Random walk





$\hat{y} = w_1 x + w_0$

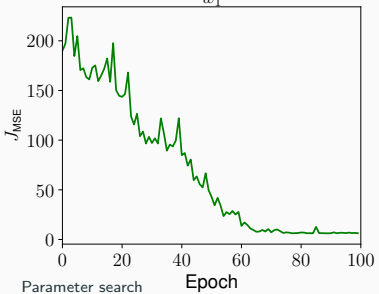
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

$\Delta w_2 \sim \mathcal{N}(\mu = 0, \sigma^2 = 0.01)$

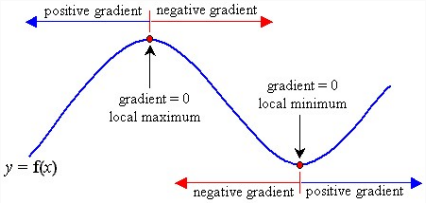
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.20$



Review: Gradient

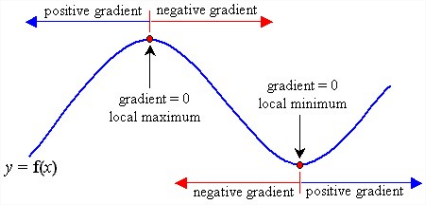
Given a graph of a function, $f(x)$,



Review: Gradient

Given a graph of a function, $f(x)$,

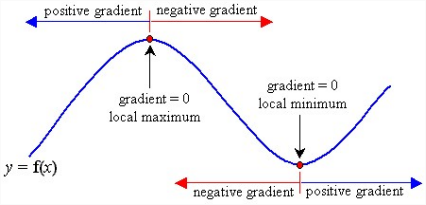
- **Local maximum** is the point where $f(x)$ changes from increasing to decreasing.



Review: Gradient

Given a graph of a function, $f(x)$,

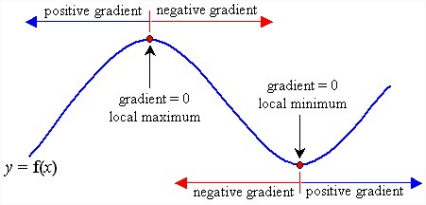
- **Local maximum** is the point where $f(x)$ changes from increasing to decreasing.
- **Local minimum** is the point where $f(x)$ changes from decreasing to increasing.



Review: Gradient

Given a graph of a function, $f(x)$,

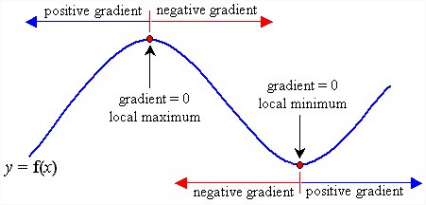
- **Local maximum** is the point where $f(x)$ changes from increasing to decreasing.
- **Local minimum** is the point where $f(x)$ changes from decreasing to increasing.
- At the local maximum/minimum, the gradient is 0.



Review: Gradient

Given a graph of a function, $f(x)$,

- **Local maximum** is the point where $f(x)$ changes from increasing to decreasing.
- **Local minimum** is the point where $f(x)$ changes from decreasing to increasing.
- At the local maximum/minimum, the gradient is 0.

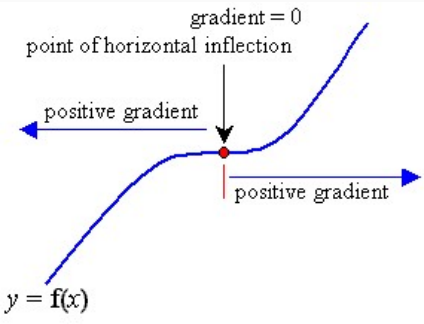


The gradient of $f(x)$ is equal to the derivative of $f(x)$: $f'(x) = \frac{\partial f(x)}{\partial x}$

Review: Gradient

However, gradient at inflection point is 0.

To find "true" local minimum, we need to look at the gradient before and after the point.



Steepest gradient descent optimisation

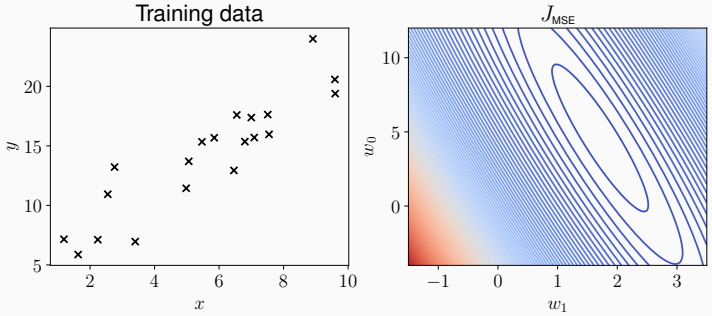
Steepest gradient descent is an informed parameter search that works only with a **continuous** loss functions. By evaluating partial derivatives of the loss function with respect to the parameters of the model, you can derive a gradient vector, direction in the parameters space pointing from w to the steepest incline of the loss surface. Then, simply set Δw to the negative of the gradient.

$$\Delta w = -\left[\frac{\partial J(w)}{\partial w_1} \quad \frac{\partial J(w)}{\partial w_2} \quad \cdots \quad \frac{\partial J(w)}{\partial w_W}\right],$$

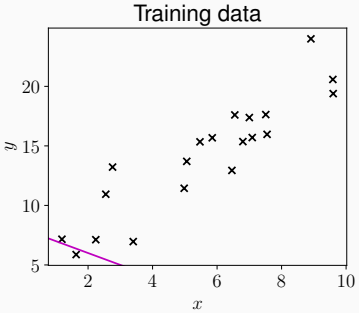
where W is the total number of parameters in the hypothesis $h(x, w)$.

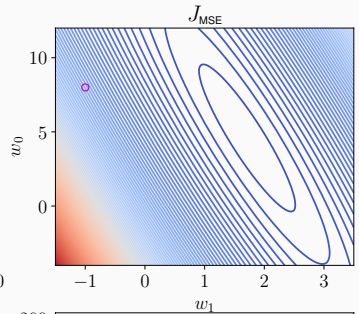
$J(w)$ is guaranteed to decrease if α is small enough!

Example: Steepest gradient descent linear regression ($\alpha = 0.02$)



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

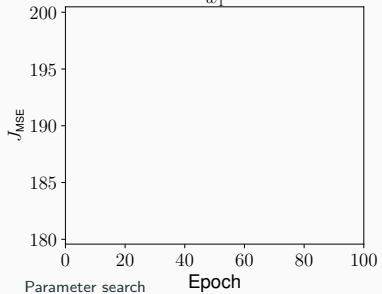
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

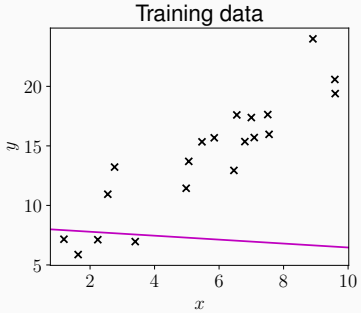
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

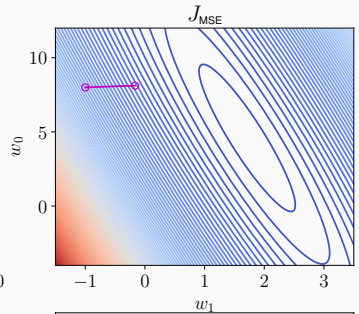
$w_1 = -1.0, w_0 = 8$

$J_{\text{MSE}} = 190.03$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

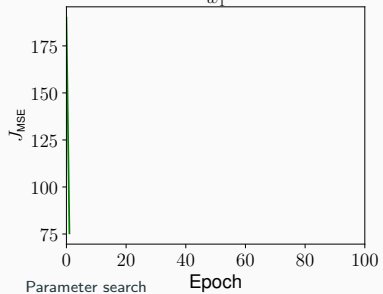
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

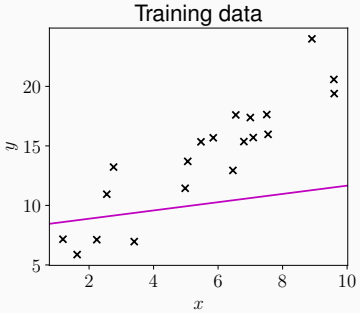
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

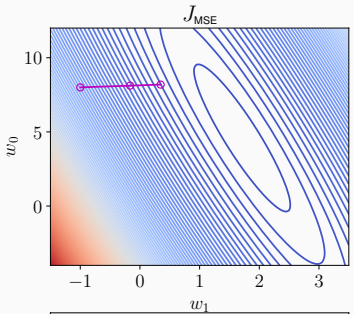
$w_1 = -0.2, w_0 = 8$

$J_{\text{MSE}} = 75.36$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

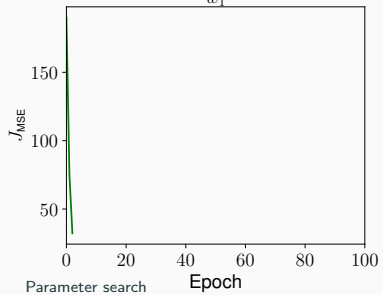
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

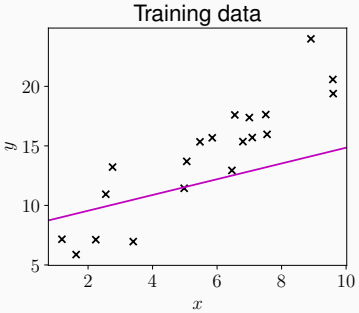
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

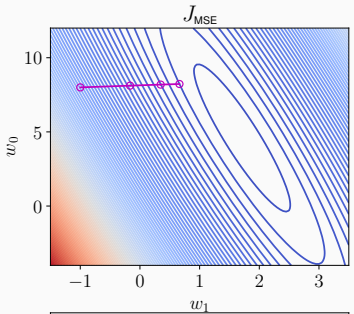
$w_1 = 0.3, w_0 = 8$

$J_{\text{MSE}} = 32.21$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

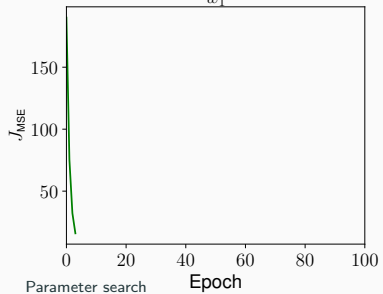
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

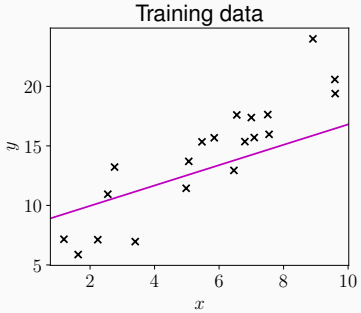
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

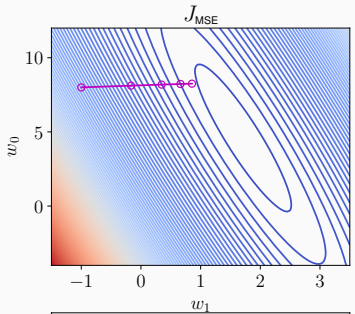
$w_1 = 0.7, w_0 = 8$

$J_{\text{MSE}} = 15.96$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

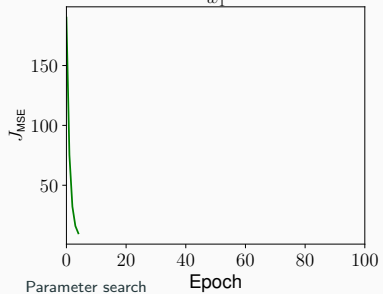
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

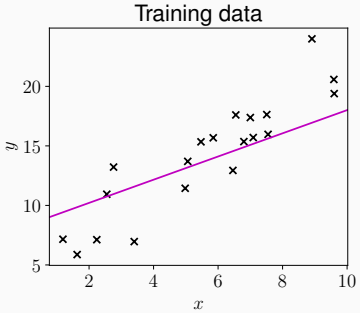
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

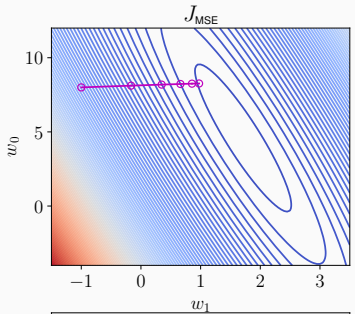
$w_1 = 0.9, w_0 = 8$

$J_{\text{MSE}} = 9.84$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

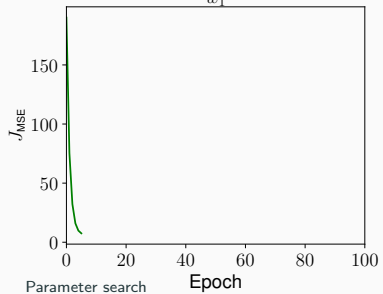
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

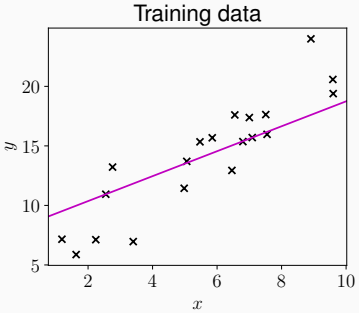
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

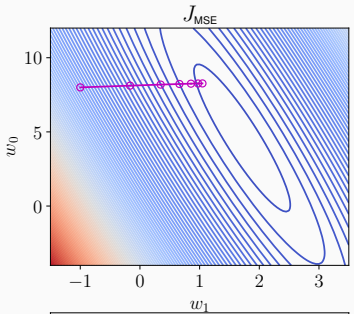
$w_1 = 1.0, w_0 = 8$

$J_{\text{MSE}} = 7.54$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

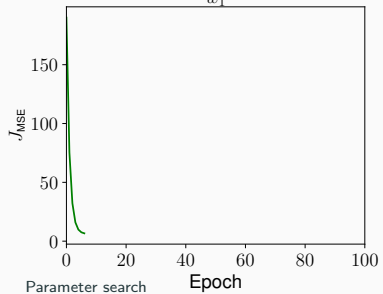
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

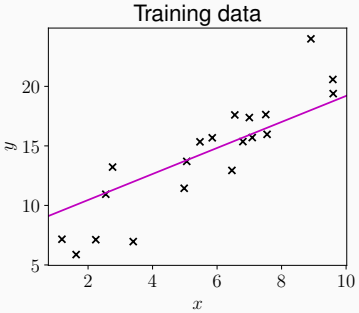
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

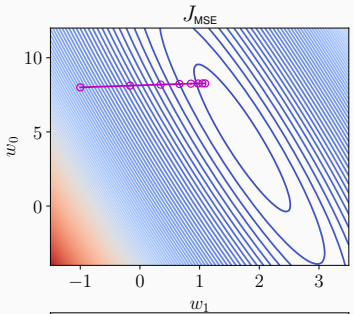
$w_1 = 1.0, w_0 = 8$

$J_{\text{MSE}} = 6.66$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

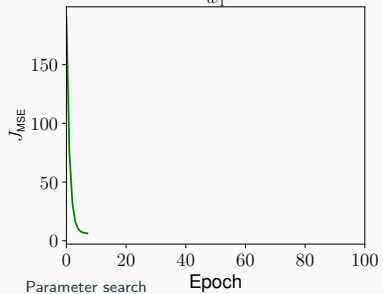
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

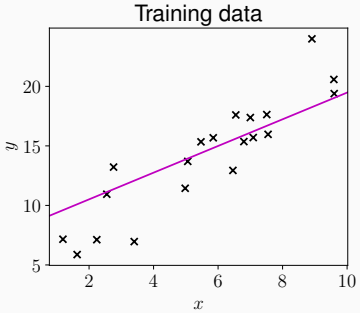
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

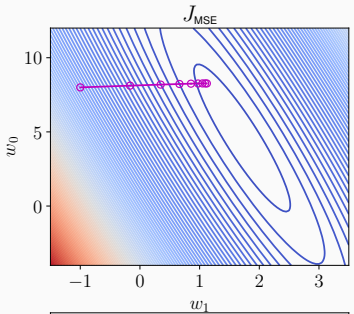
$w_1 = 1.1, w_0 = 8$

$J_{\text{MSE}} = 6.33$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

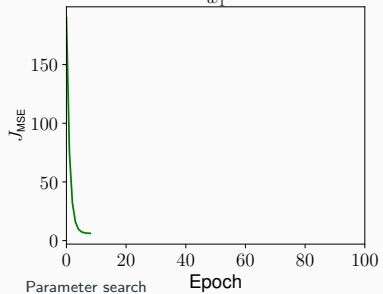
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

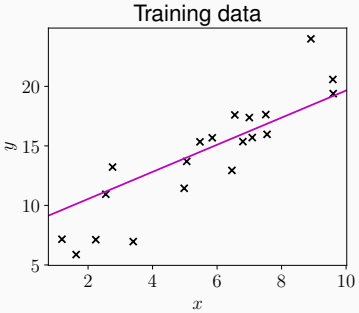
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

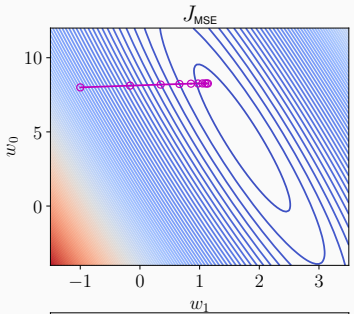
$w_1 = 1.1, w_0 = 8$

$J_{\text{MSE}} = 6.20$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

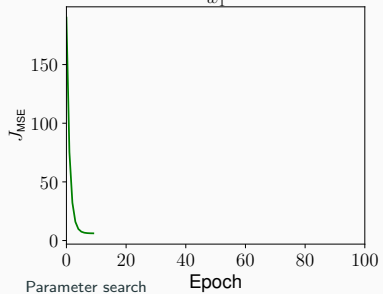
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

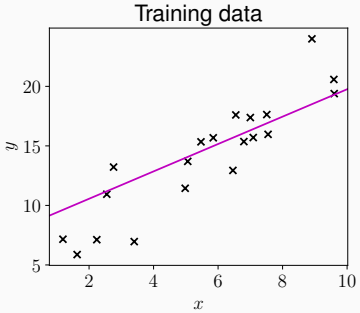
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

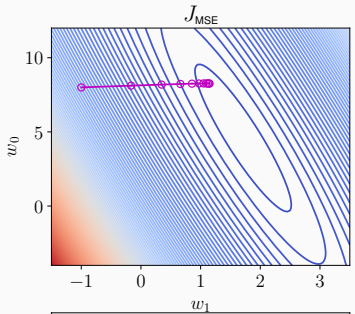
$w_1 = 1.1, w_0 = 8$

$J_{\text{MSE}} = 6.15$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

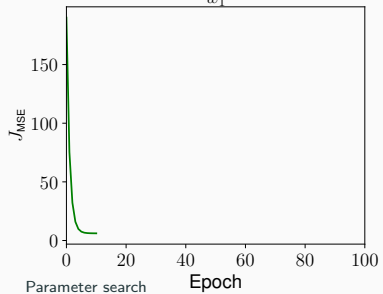
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

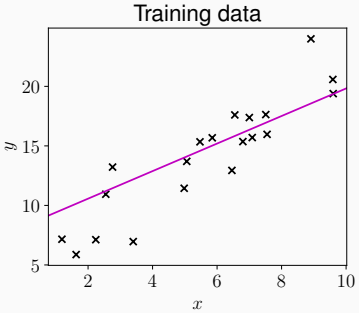
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

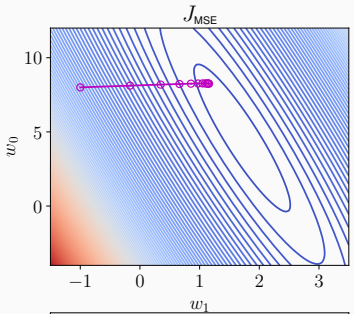
$w_1 = 1.1, w_0 = 8$

$J_{\text{MSE}} = 6.12$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

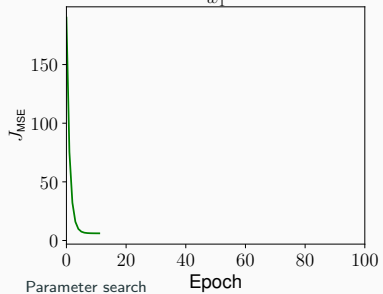
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

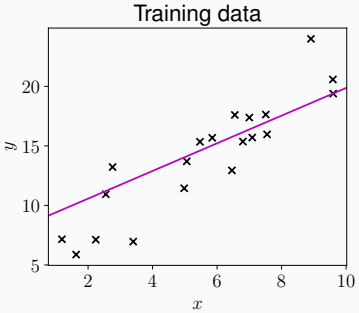
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

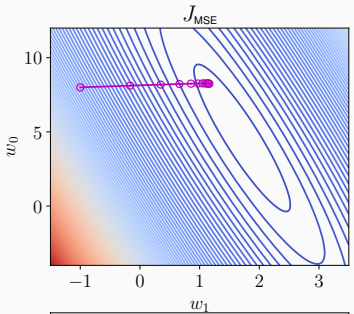
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.11$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

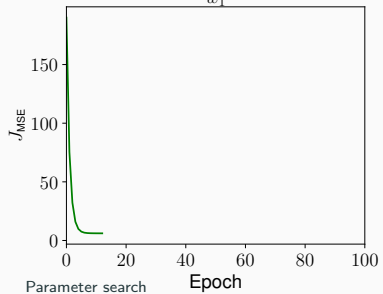
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

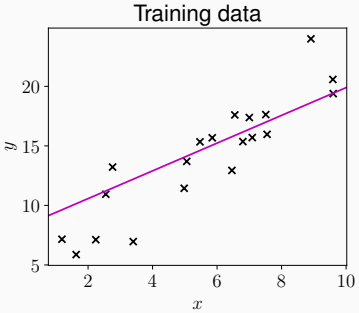
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

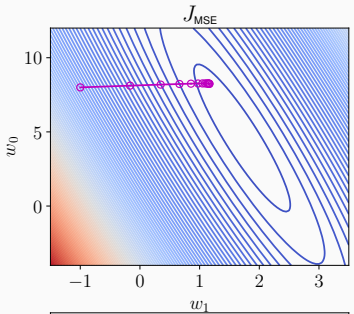
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.10$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

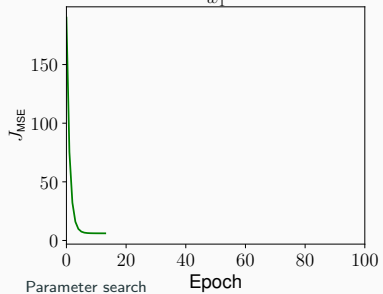
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

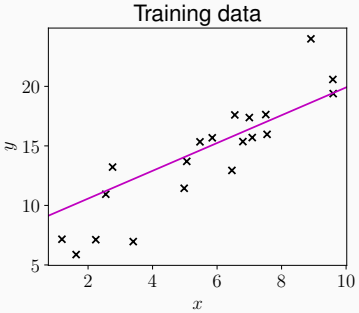
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

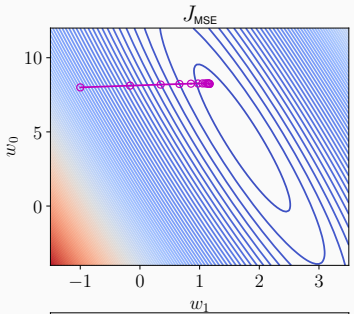
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.09$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

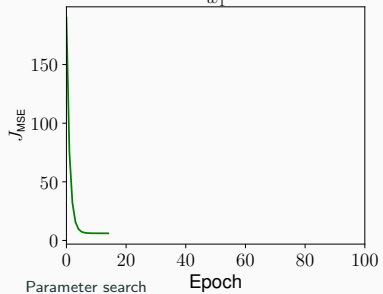
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

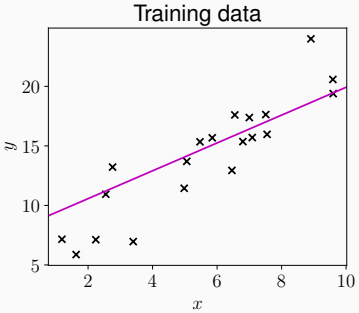
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

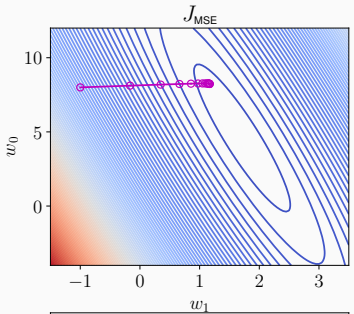
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.08$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

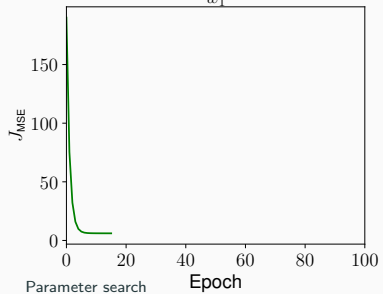
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

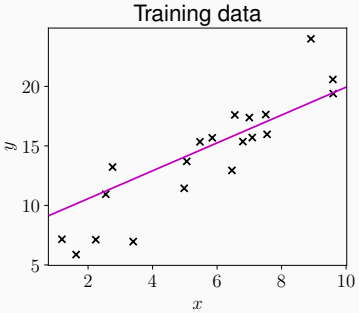
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

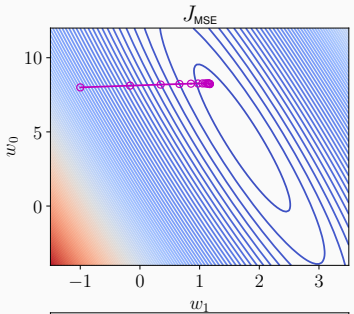
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.07$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

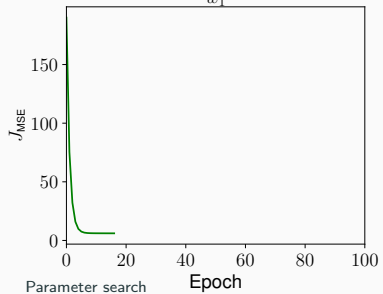
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

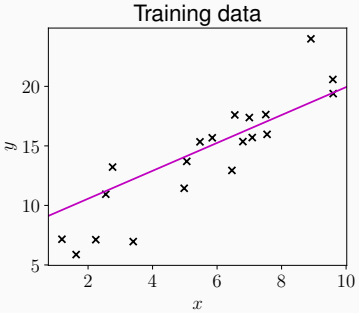
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

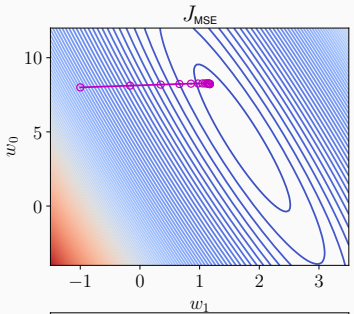
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.07$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

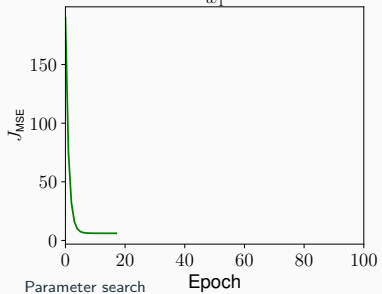
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

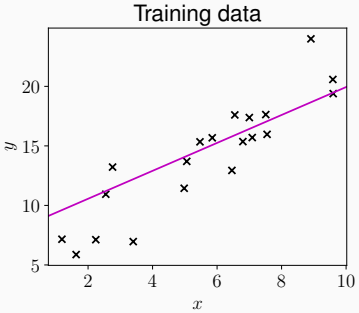
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

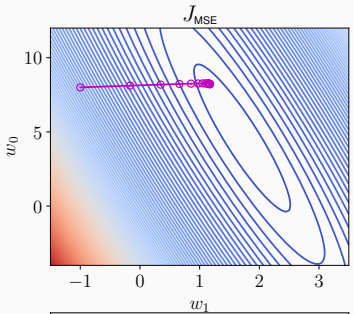
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.06$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

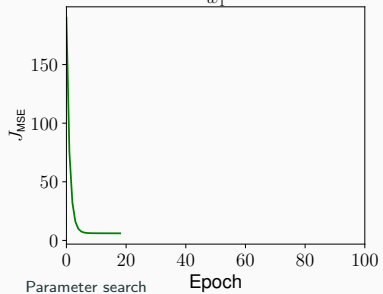
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

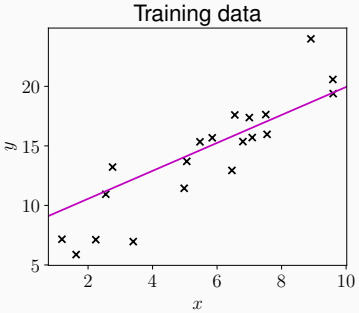
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

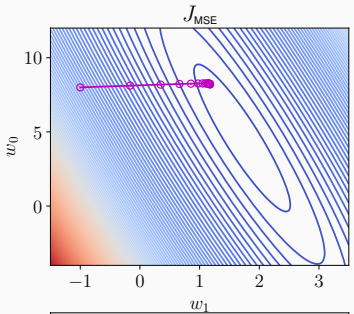
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.05$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

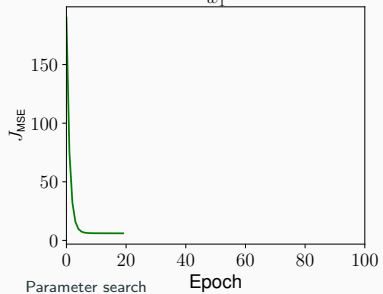
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

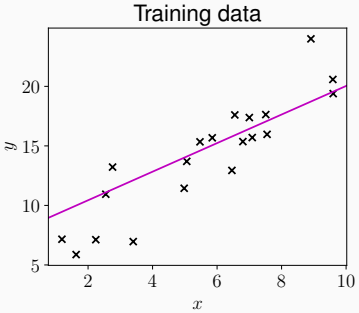
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

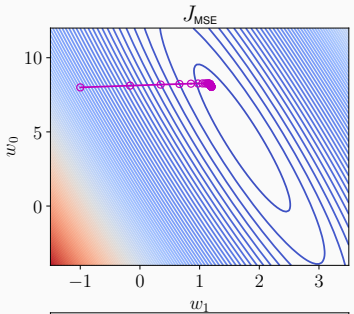
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 6.04$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

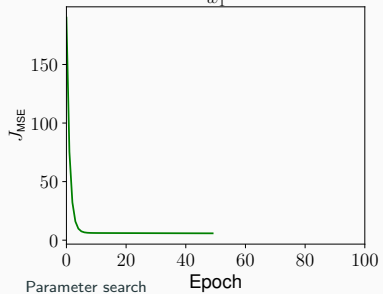
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

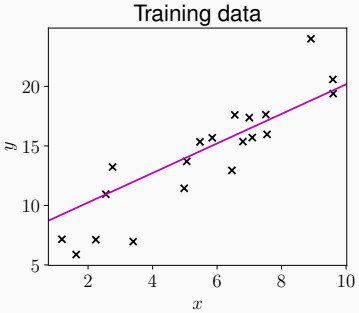
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

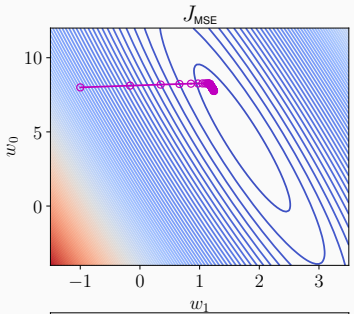
$w_1 = 1.2, w_0 = 8$

$J_{\text{MSE}} = 5.84$



Example: Steepest gradient descent linear regression ($\alpha = 0.02$)





$\hat{y} = w_1 x + w_0$

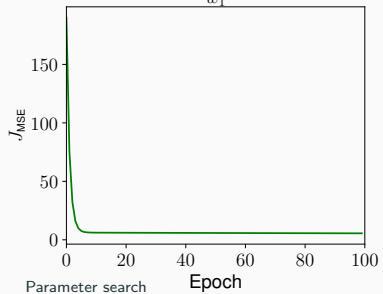
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

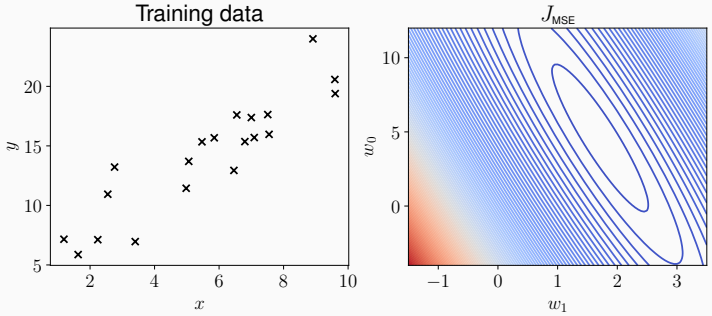
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

$w_1 = 1.2, w_0 = 8$

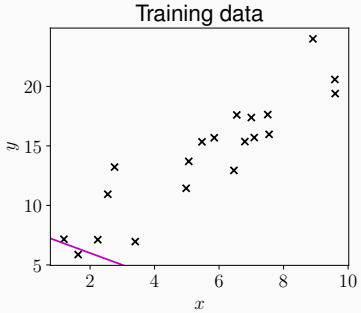
$J_{\text{MSE}} = 5.54$

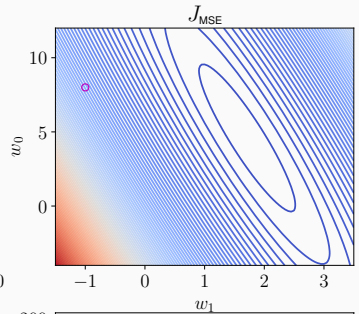


Example: Steepest gradient descent linear regression ($\alpha = 0.1$)



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

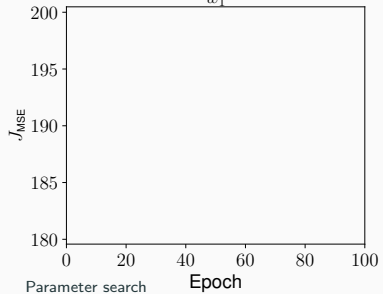
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

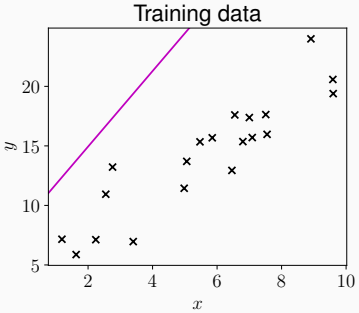
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

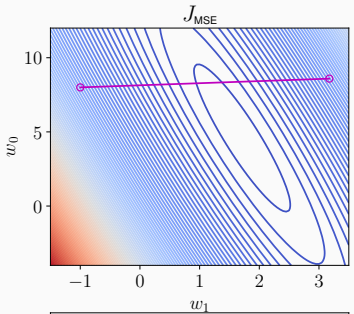
$w_1 = -1.0, w_0 = 8$

$J_{\text{MSE}} = 190.03$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

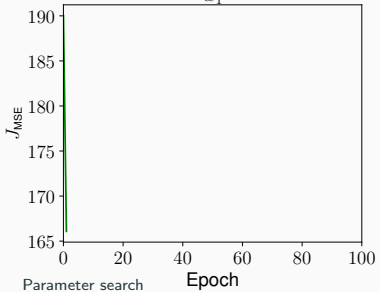
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

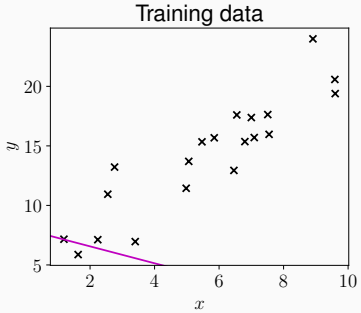
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

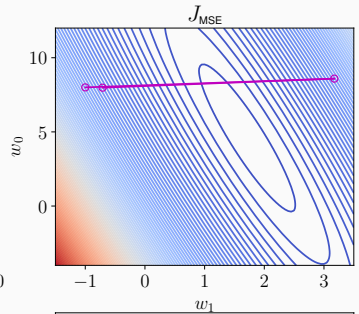
$w_1 = 3.2, w_0 = 9$

$J_{\text{MSE}} = 166.09$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

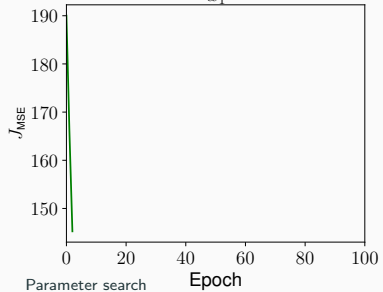
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

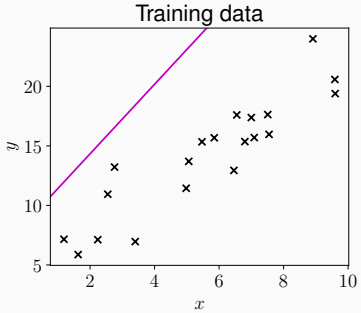
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

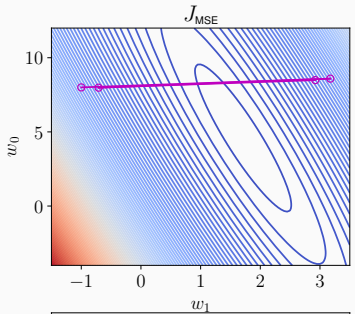
$w_1 = -0.7, w_0 = 8$

$J_{\text{MSE}} = 145.25$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

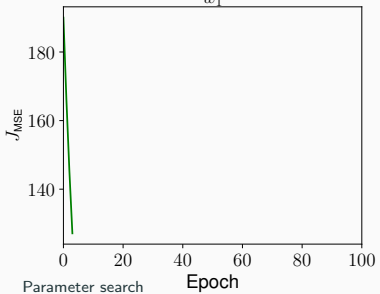
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

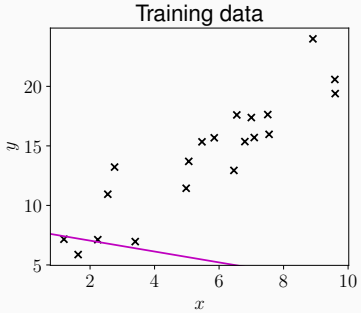
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

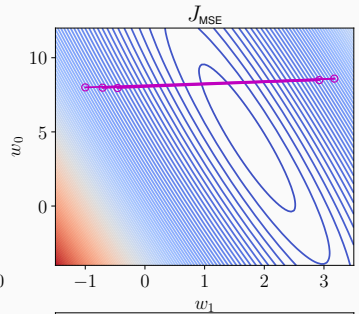
$w_1 = 2.9, w_0 = 8$

$J_{\text{MSE}} = 127.13$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

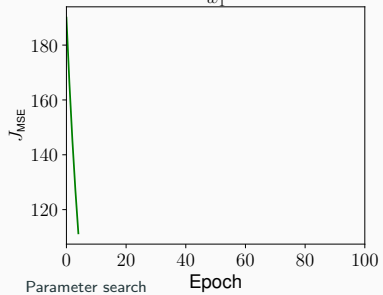
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

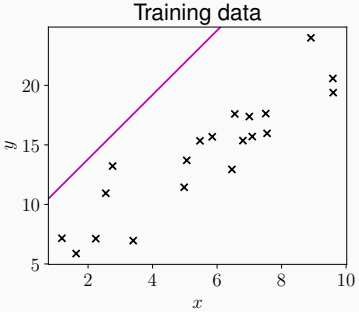
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

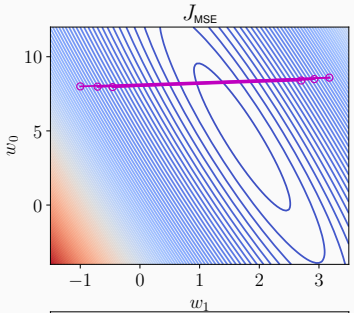
$w_1 = -0.5, w_0 = 8$

$J_{\text{MSE}} = 111.35$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

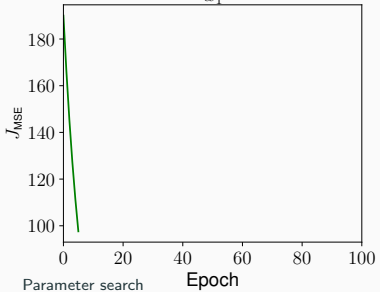
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

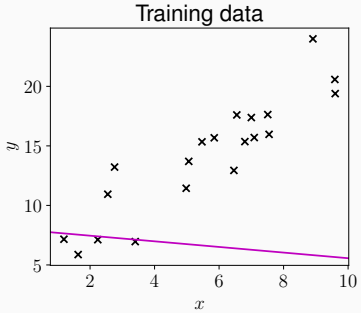
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

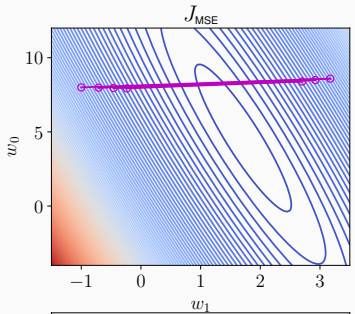
$w_1 = 2.7, w_0 = 8$

$J_{\text{MSE}} = 97.62$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

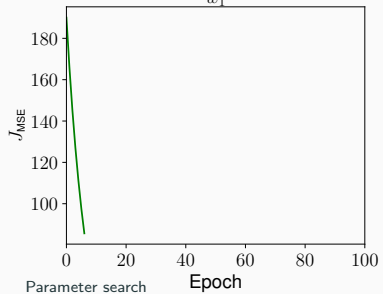
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

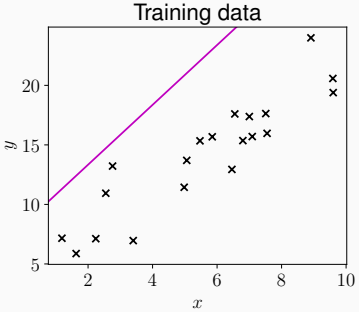
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

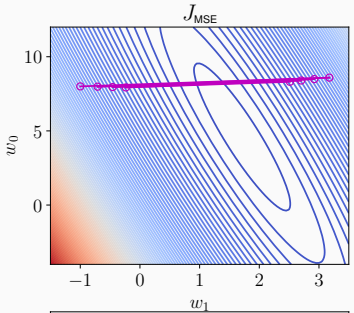
$w_1 = -0.2, w_0 = 8$

$J_{\text{MSE}} = 85.68$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

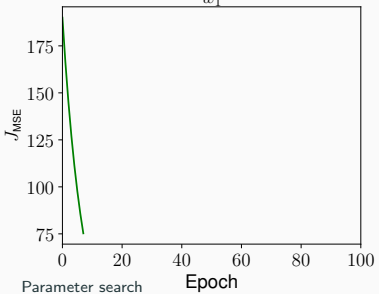
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

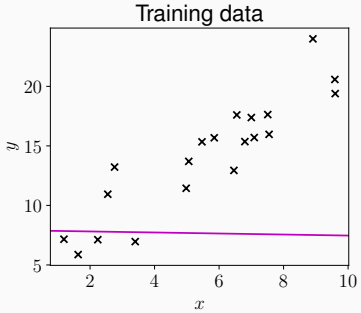
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

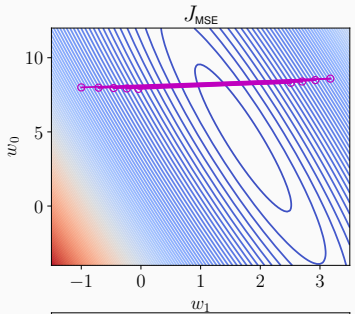
$w_1 = 2.5, w_0 = 8$

$J_{\text{MSE}} = 75.28$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

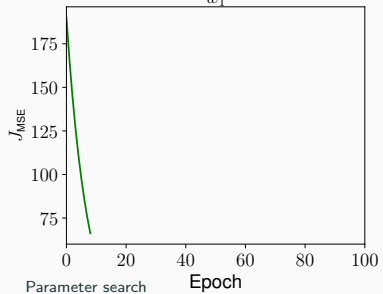
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

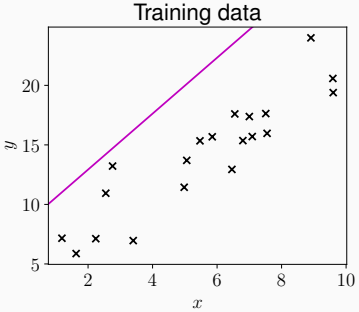
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

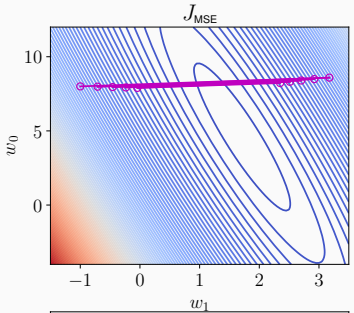
$w_1 = -0.0, w_0 = 8$

$J_{\text{MSE}} = 66.23$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

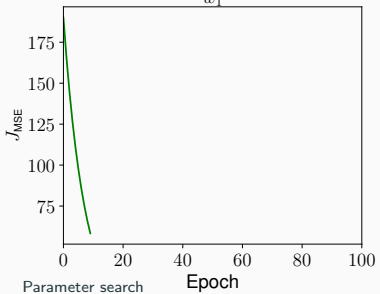
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

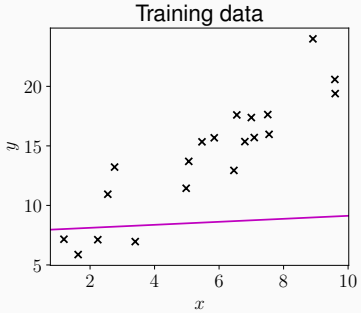
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

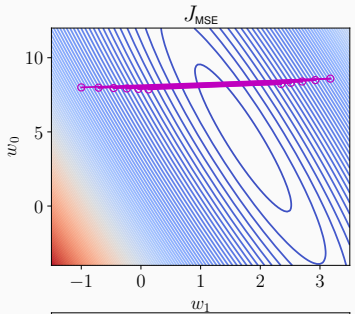
$w_1 = 2.3, w_0 = 8$

$J_{\text{MSE}} = 58.35$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

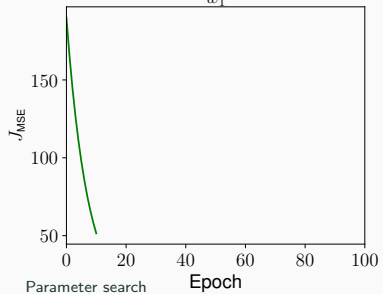
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

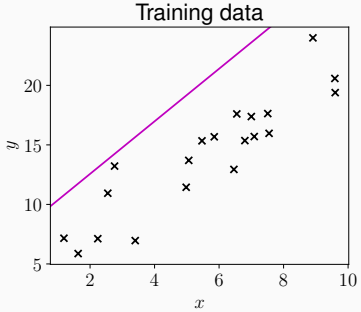
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

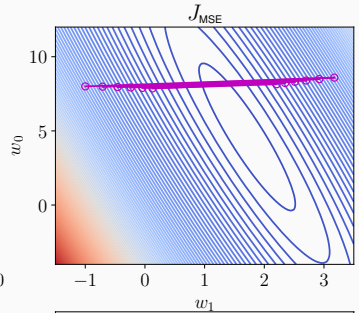
$w_1 = 0.1, w_0 = 8$

$J_{\text{MSE}} = 51.49$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

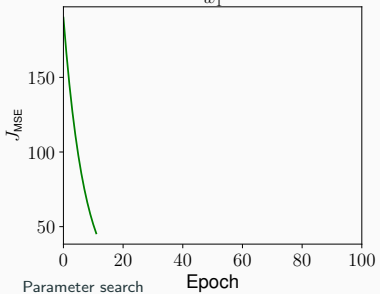
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

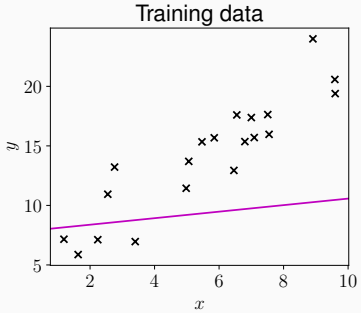
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

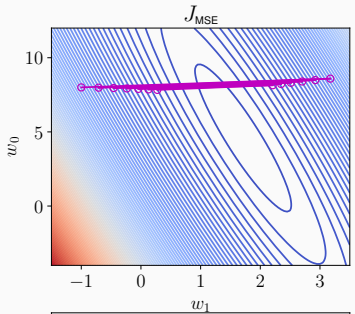
$w_1 = 2.2, w_0 = 8$

$J_{\text{MSE}} = 45.52$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

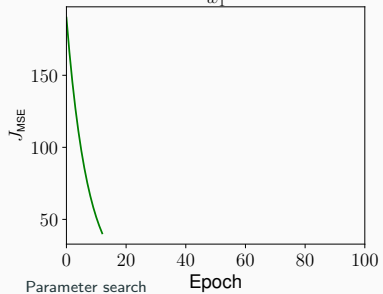
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

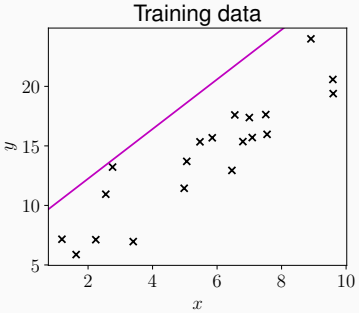
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

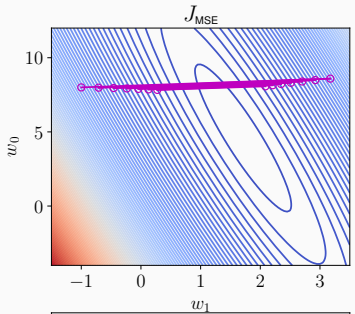
$w_1 = 0.3, w_0 = 8$

$J_{\text{MSE}} = 40.33$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

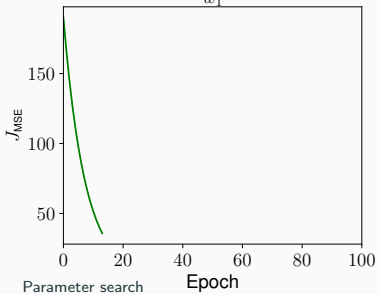
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

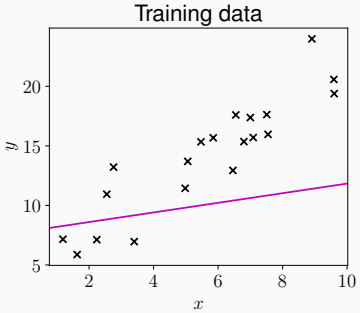
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

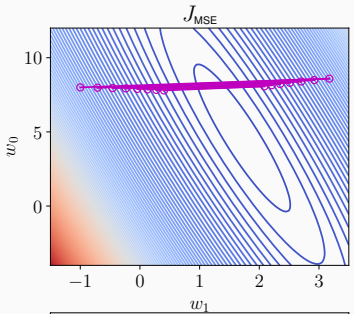
$w_1 = 2.1, w_0 = 8$

$J_{\text{MSE}} = 35.80$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

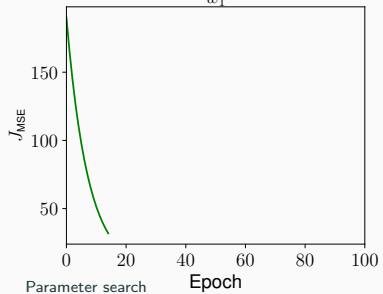
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

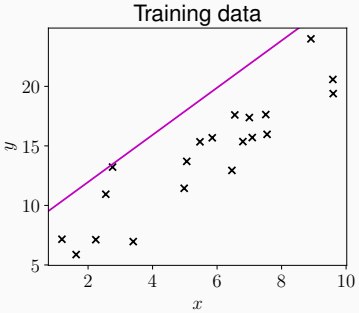
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

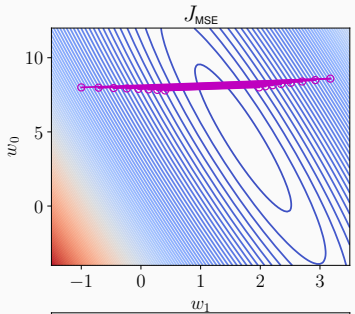
$w_1 = 0.4, w_0 = 8$

$J_{\text{MSE}} = 31.86$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

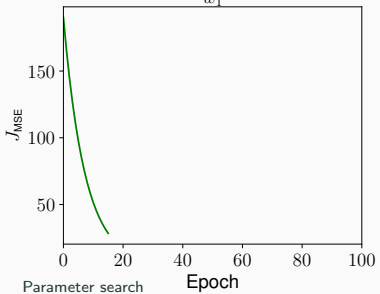
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

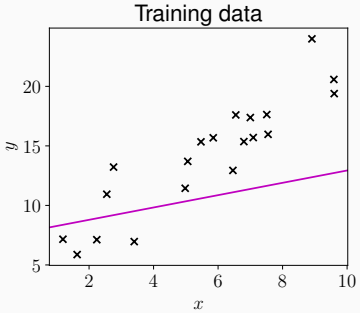
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

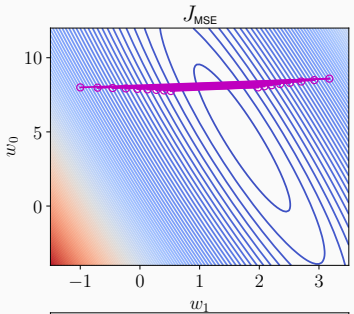
$w_1 = 2.0, w_0 = 8$

$J_{\text{MSE}} = 28.43$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

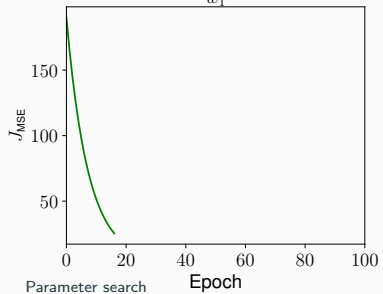
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

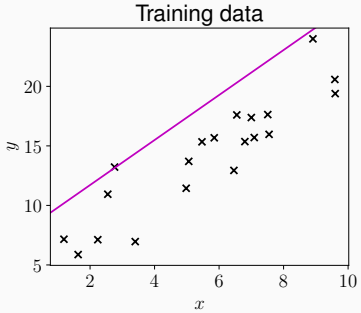
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

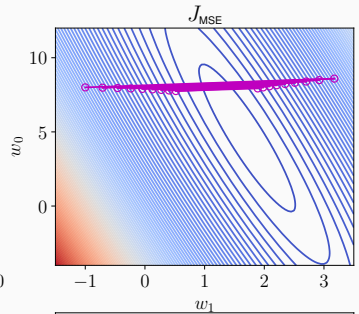
$w_1 = 0.5, w_0 = 8$

$J_{\text{MSE}} = 25.44$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

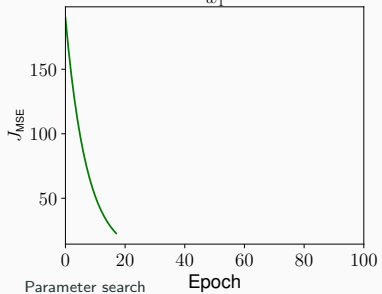
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

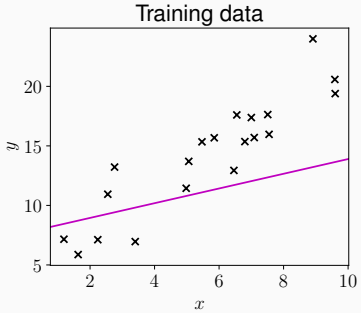
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

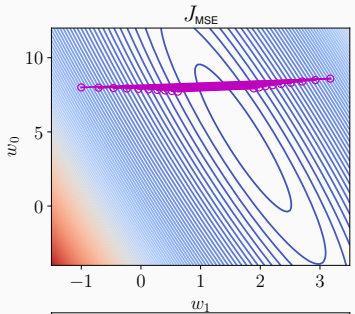
$w_1 = 1.9, w_0 = 8$

$J_{\text{MSE}} = 22.84$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

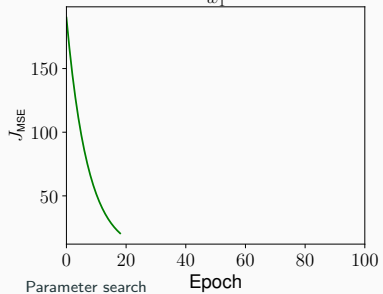
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

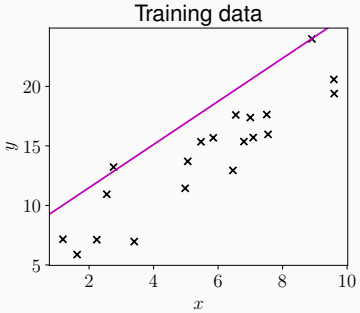
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

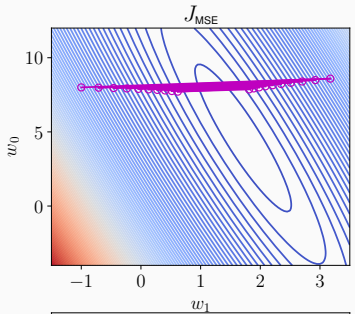
$w_1 = 0.6, w_0 = 8$

$J_{\text{MSE}} = 20.57$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

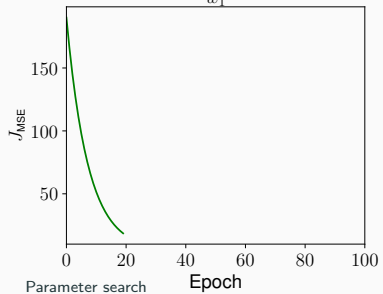
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

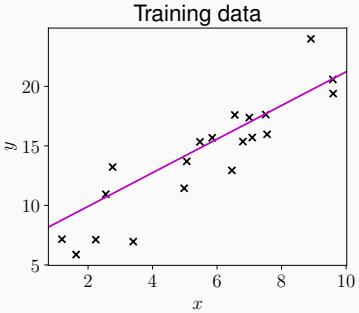
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

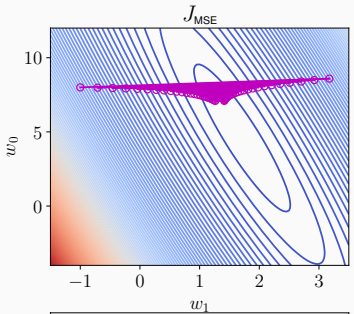
$w_1 = 1.8, w_0 = 8$

$J_{\text{MSE}} = 18.59$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

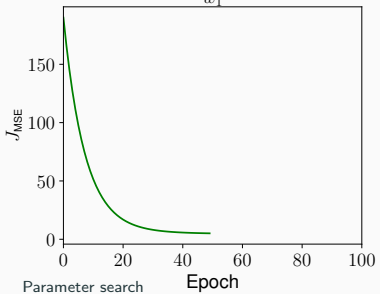
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

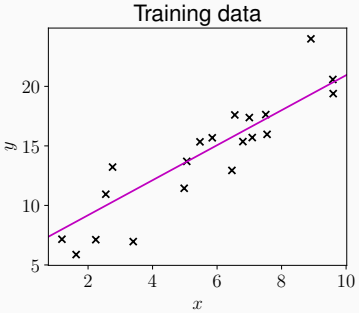
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

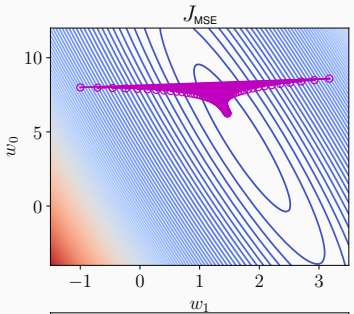
$w_1 = 1.4, w_0 = 7$

$J_{\text{MSE}} = 5.09$



Example: Steepest gradient descent linear regression ($\alpha = 0.1$)





$\hat{y} = w_1 x + w_0$

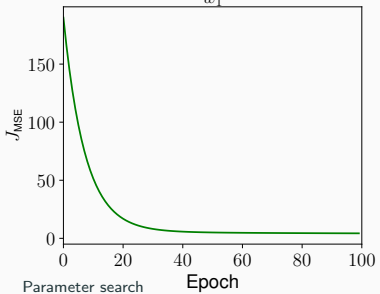
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

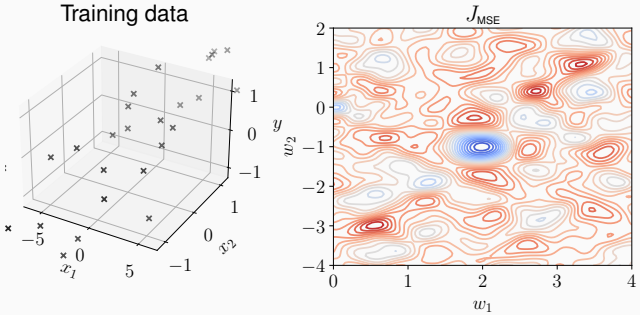
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

$w_1 = 1.5, w_0 = 6$

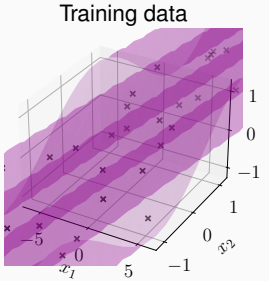
$J_{\text{MSE}} = 4.30$

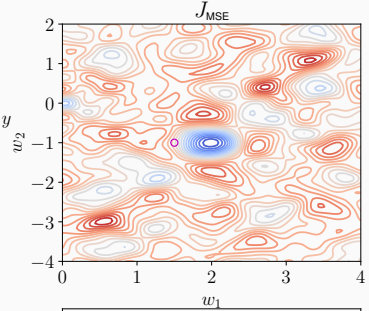


Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

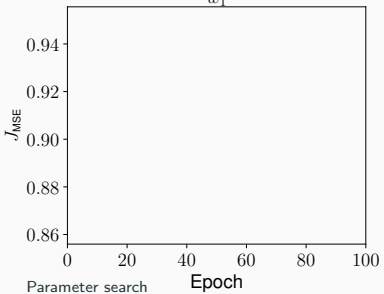
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

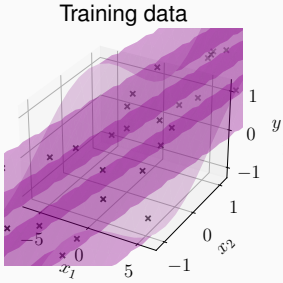
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

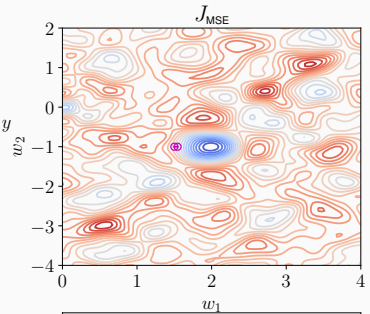
$w_1 = 1.5, w_2 = -1$

$J_{\text{MSE}} = 0.91$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

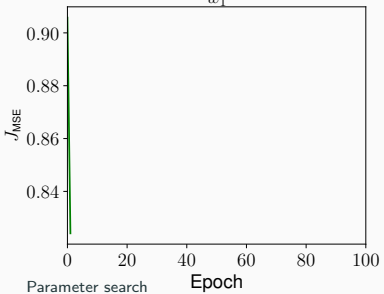
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

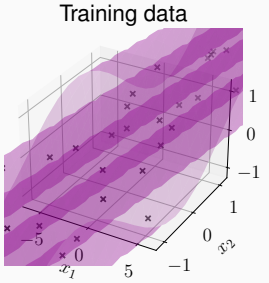
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

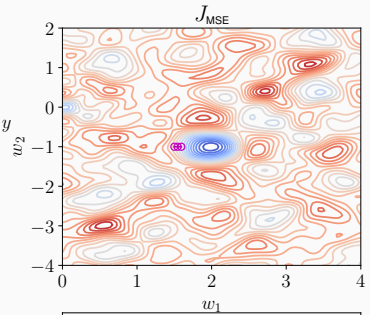
$w_1 = 1.5, w_2 = -1$

$J_{\text{MSE}} = 0.82$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

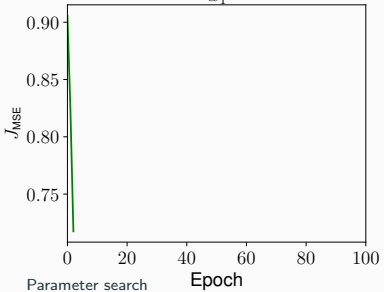
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

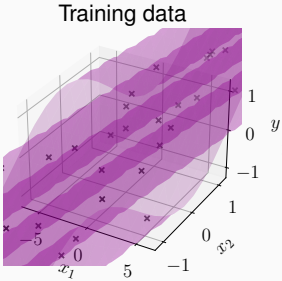
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

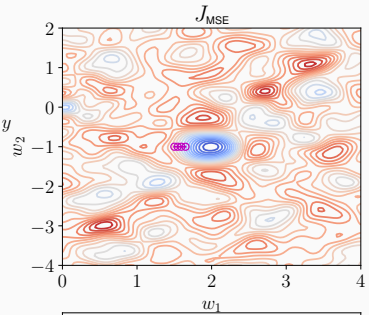
$w_1 = 1.6, w_2 = -1$

$J_{\text{MSE}} = 0.72$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

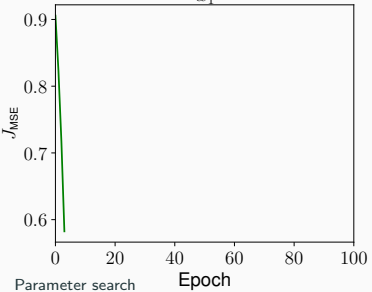
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

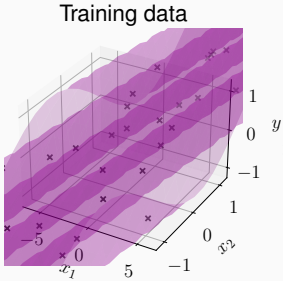
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

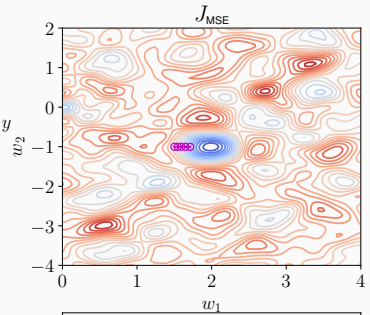
$w_1 = 1.7, w_2 = -1$

$J_{\text{MSE}} = 0.58$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

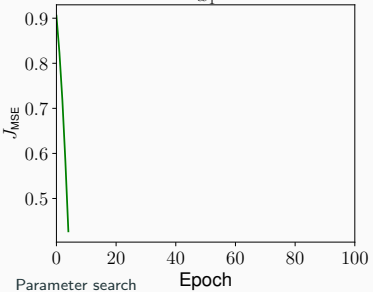
$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

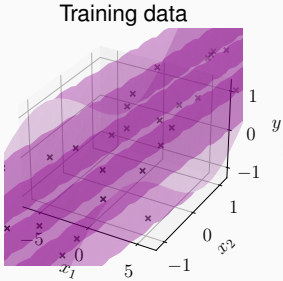
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

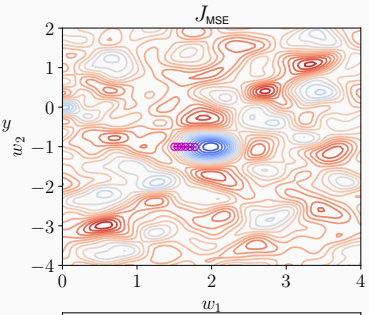
$w_1 = 1.7, w_2 = -1$

$J_{\text{MSE}} = 0.43$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

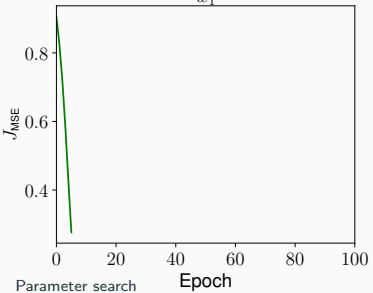
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

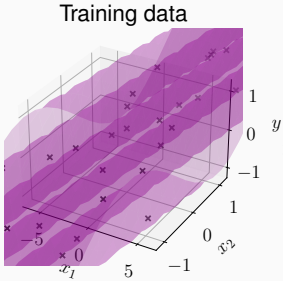
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

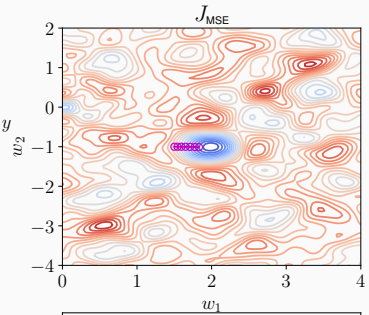
$w_1 = 1.8, w_2 = -1$

$J_{\text{MSE}} = 0.28$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

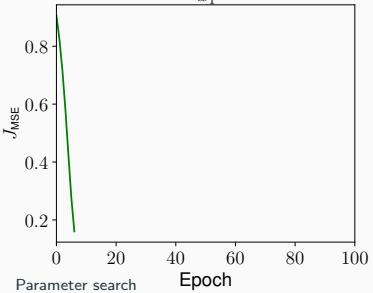
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

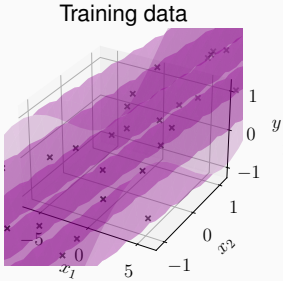
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

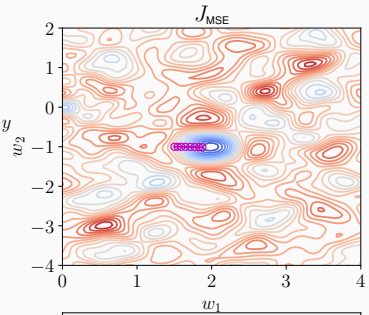
$w_1 = 1.8, w_2 = -1$

$J_{\text{MSE}} = 0.16$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

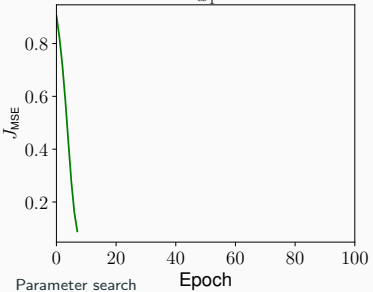
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

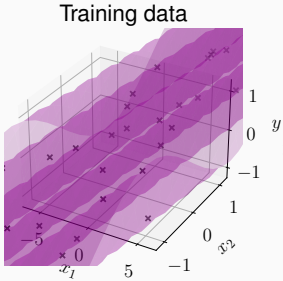
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

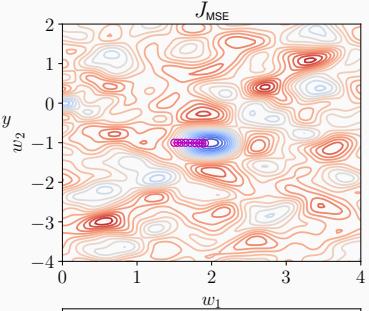
$$w_1 = 1.9, w_2 = -1$$

$$J_{\text{MSE}} = 0.09$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

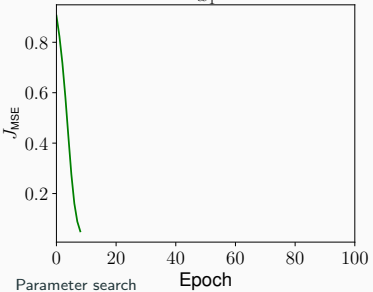
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

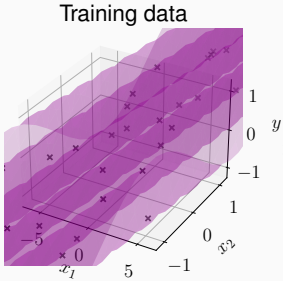
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

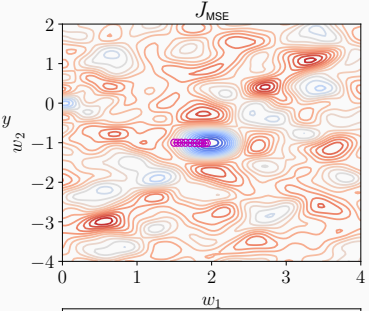
$$w_1 = 1.9, w_2 = -1$$

$$J_{\text{MSE}} = 0.05$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

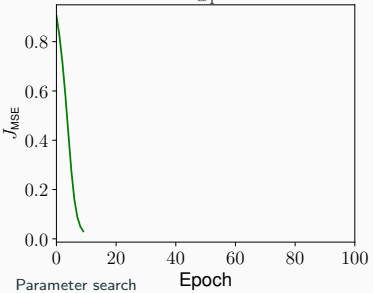
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

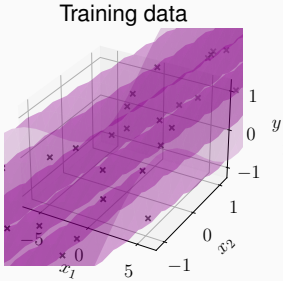
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

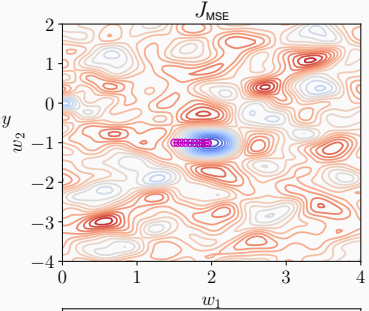
$$w_1 = 1.9, w_2 = -1$$

$$J_{\text{MSE}} = 0.03$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

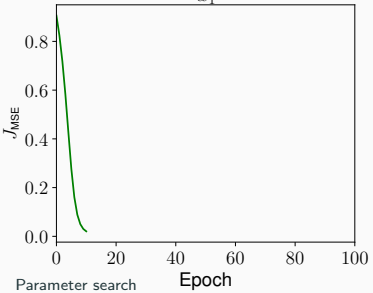
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

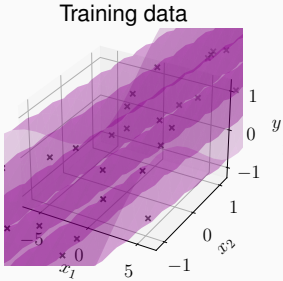
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

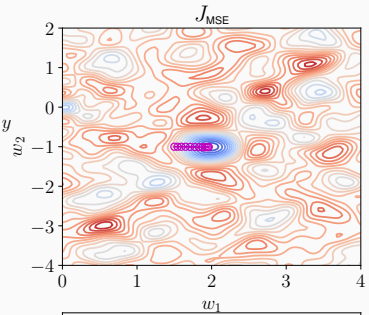
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.02$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

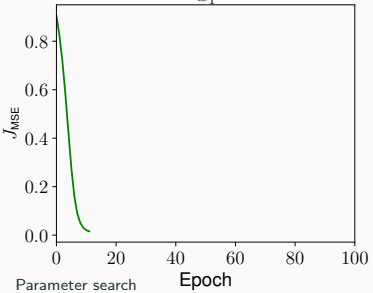
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

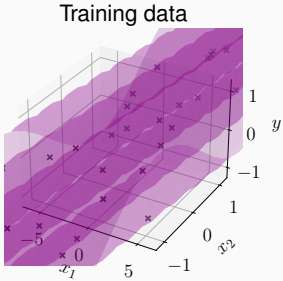
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

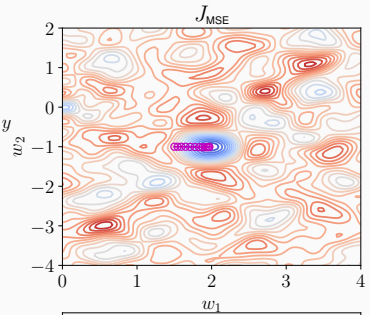
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.02$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

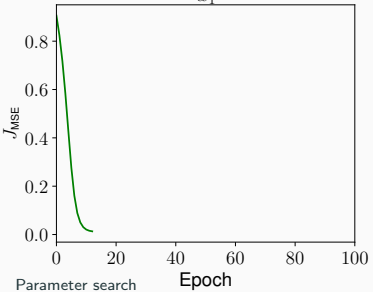
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

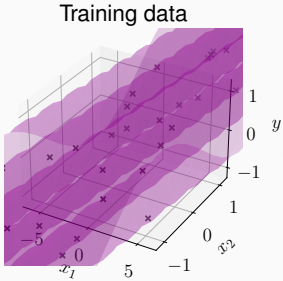
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

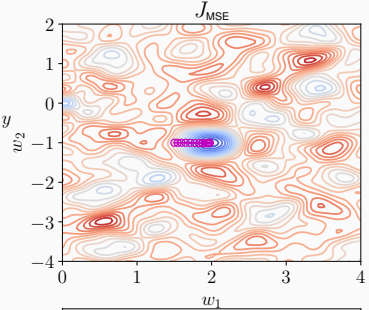
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

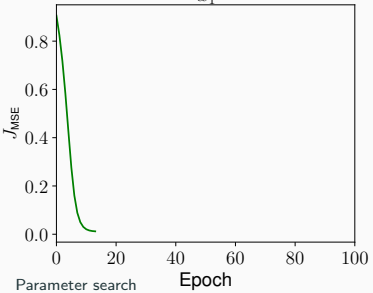
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

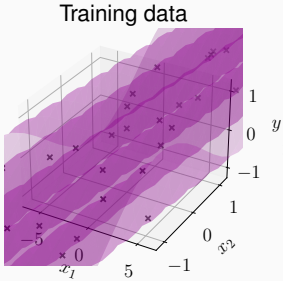
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

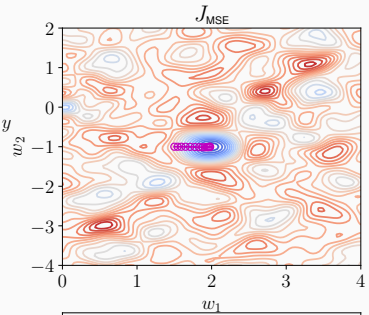
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

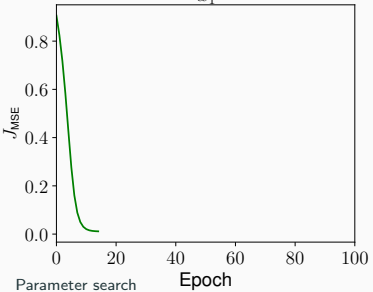
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

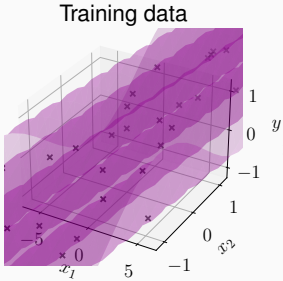
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

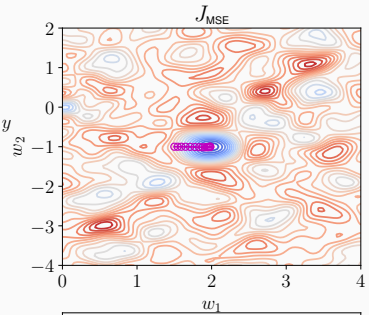
$w_1 = 2.0, w_2 = -1$

$J_{\text{MSE}} = 0.01$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

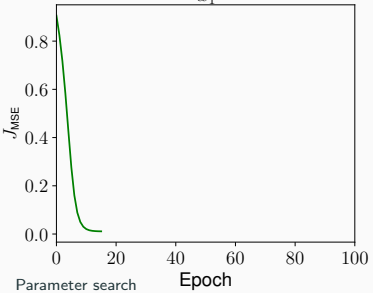
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

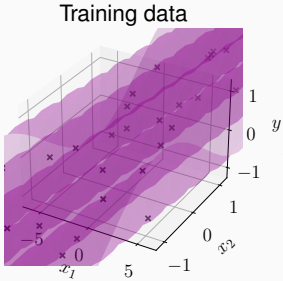
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

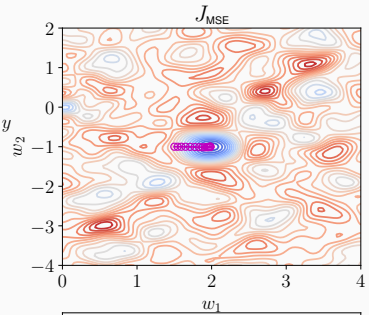
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

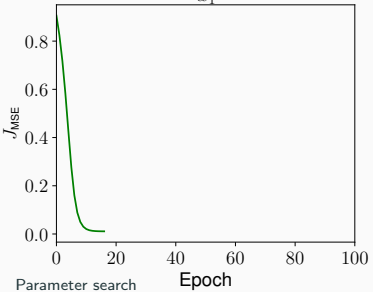
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

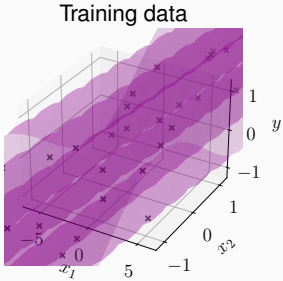
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

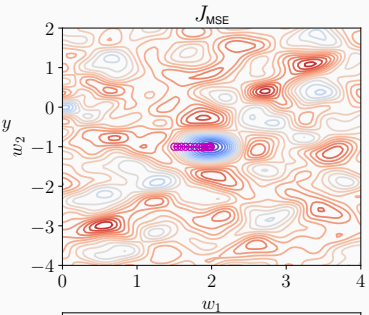
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

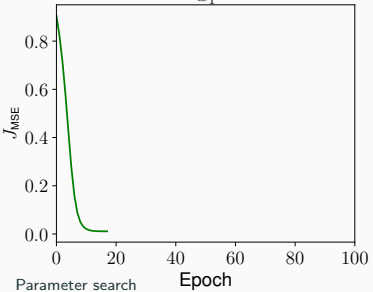
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

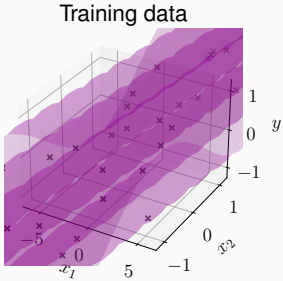
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

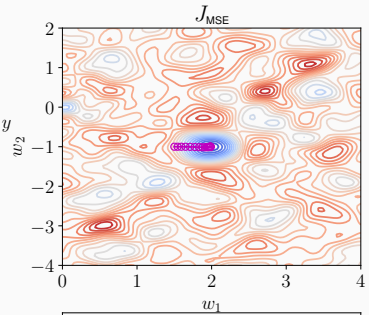
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

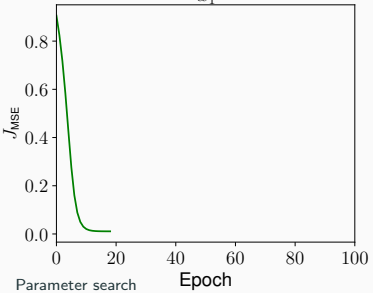
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

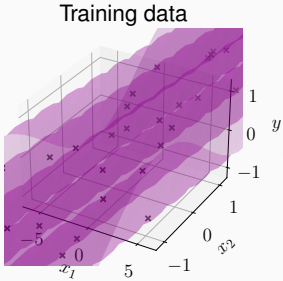
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

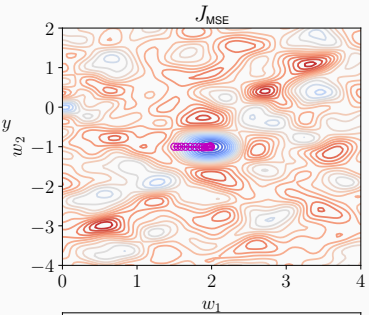
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

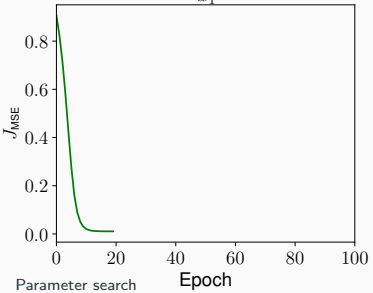
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

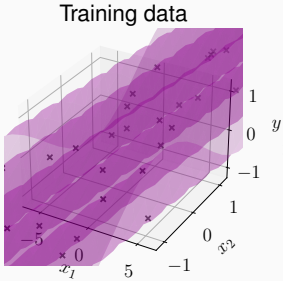
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

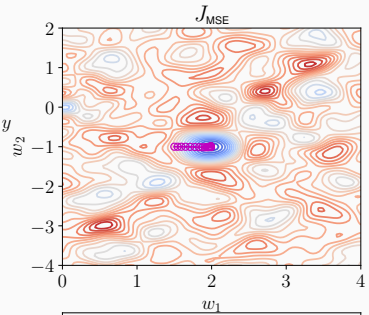
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

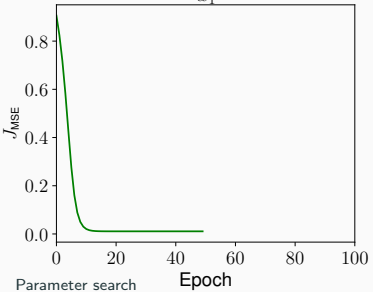
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

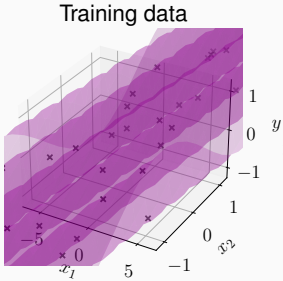
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

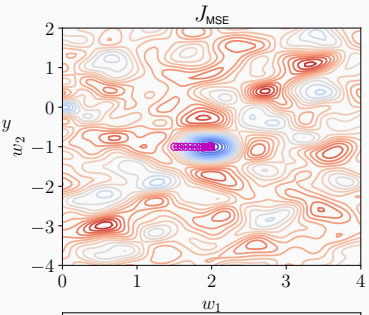
$$w_1 = 2.0, w_2 = -1$$

$$J_{\text{MSE}} = 0.01$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

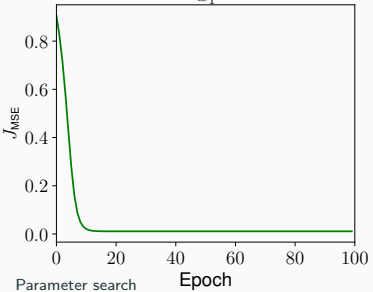
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

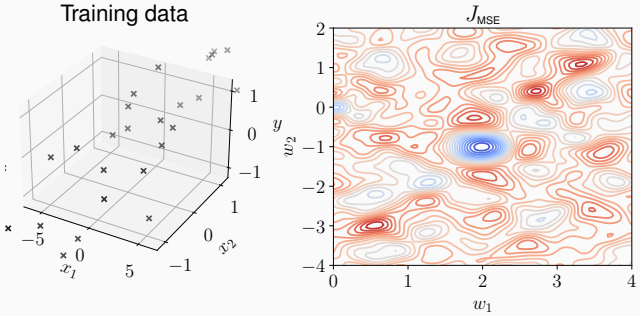
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

$$w_1 = 2.0, w_2 = -1$$

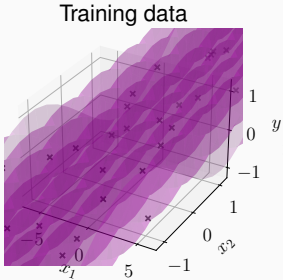
$$J_{\text{MSE}} = 0.01$$

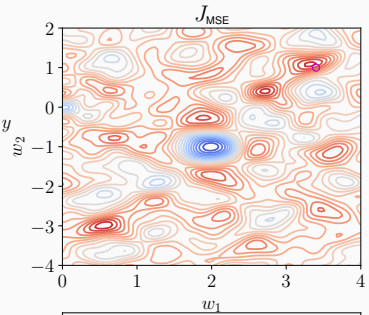


Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

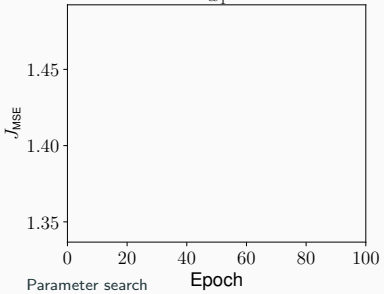
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

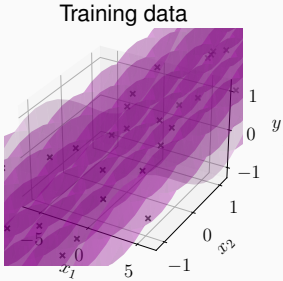
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

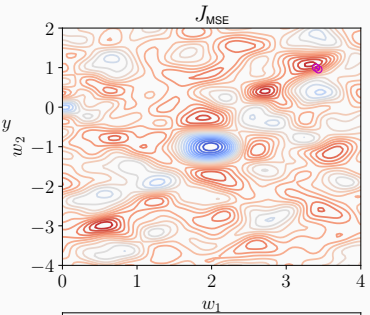
$w_1 = 3.4, w_2 = 1$

$J_{\text{MSE}} = 1.41$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

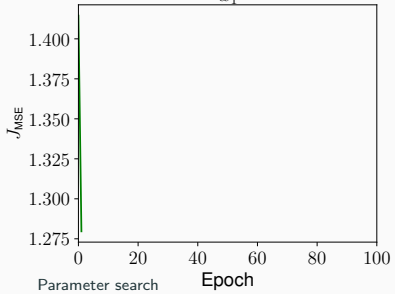
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

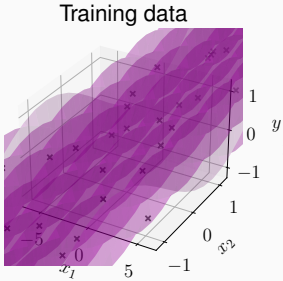
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

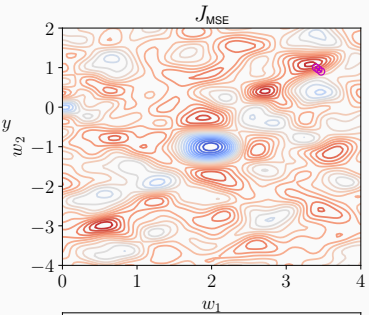
$w_1 = 3.4, w_2 = 1$

$J_{\text{MSE}} = 1.28$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

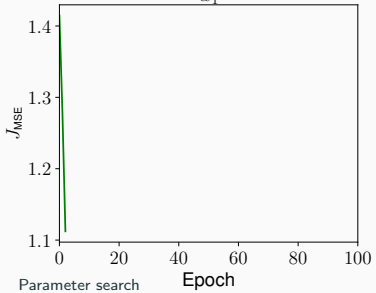
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

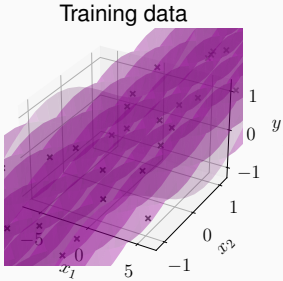
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

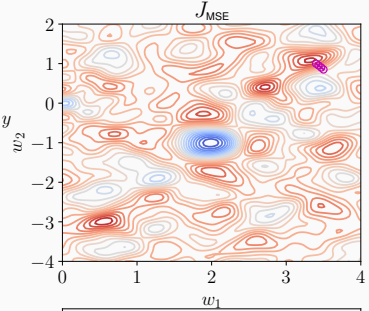
$w_1 = 3.5, w_2 = 1$

$J_{\text{MSE}} = 1.11$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

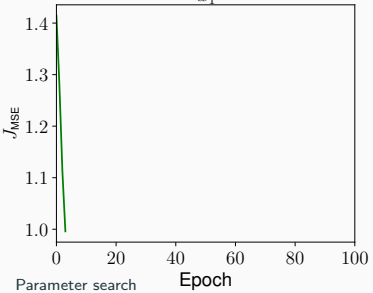
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

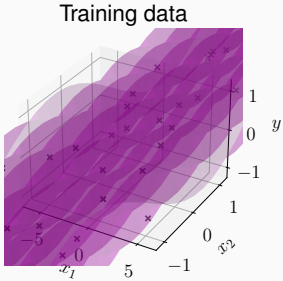
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

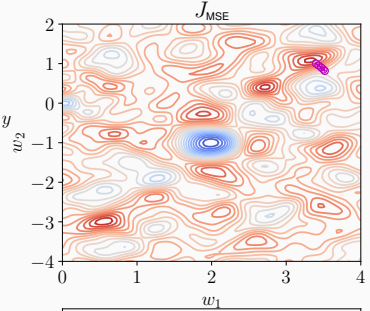
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 1.00$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

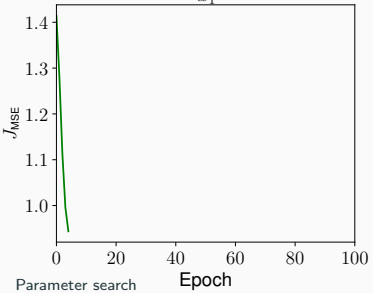
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

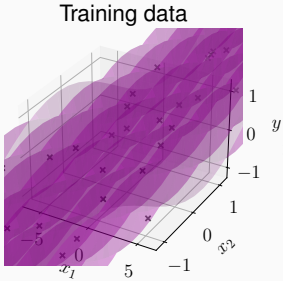
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

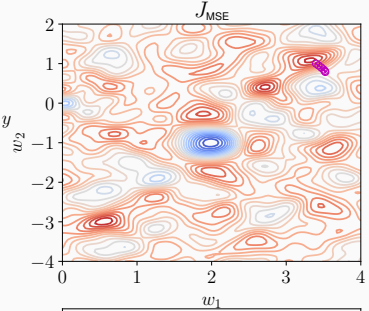
$w_1 = 3.5, w_2 = 1$

$J_{\text{MSE}} = 0.94$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

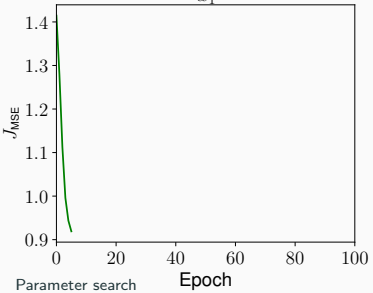
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

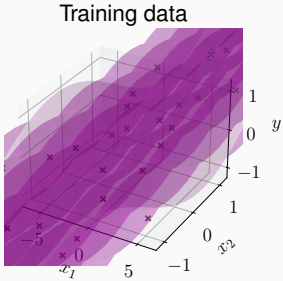
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

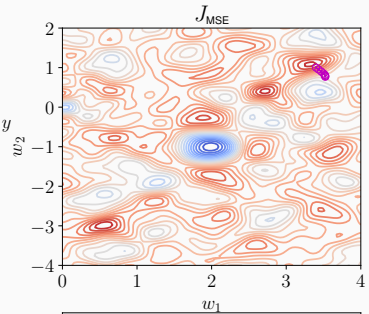
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.92$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

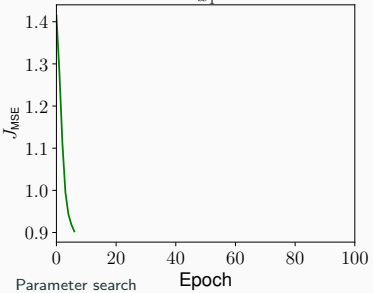
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

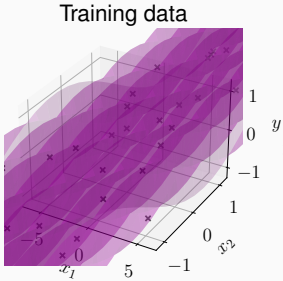
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

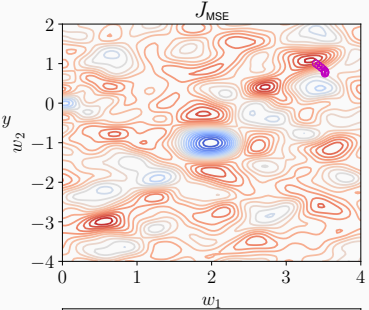
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.90$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

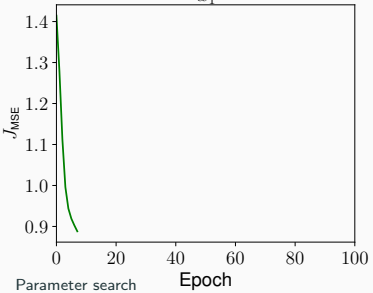
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

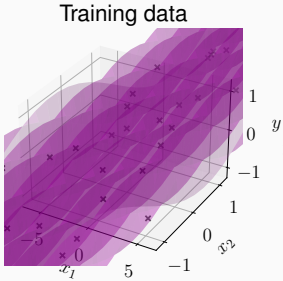
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

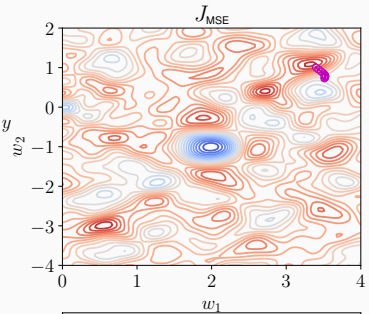
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.89$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

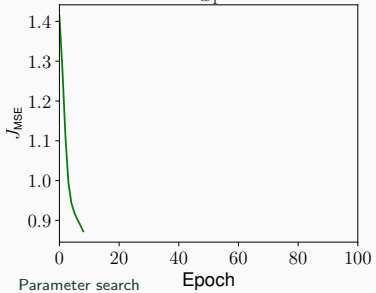
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

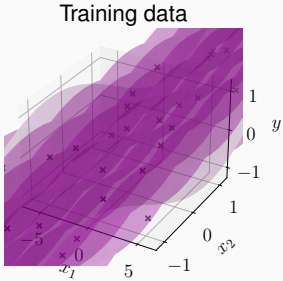
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

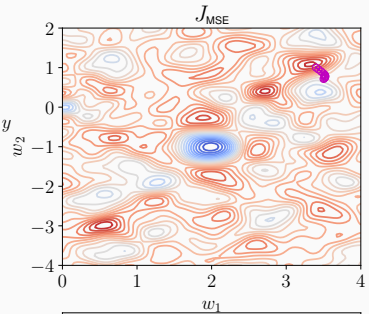
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.87$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

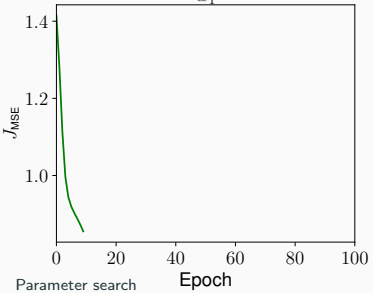
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$

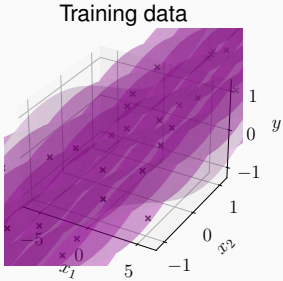
$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$

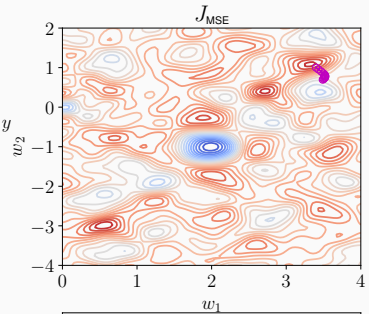
$w_1 = 3.5, w_2 = 1$

$J_{\text{MSE}} = 0.86$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

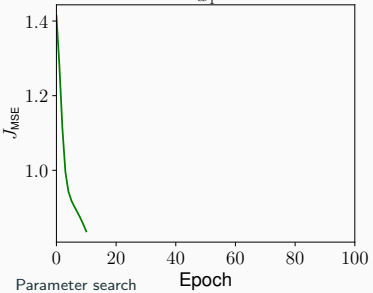
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

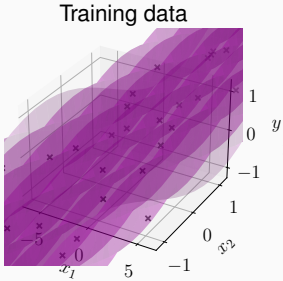
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

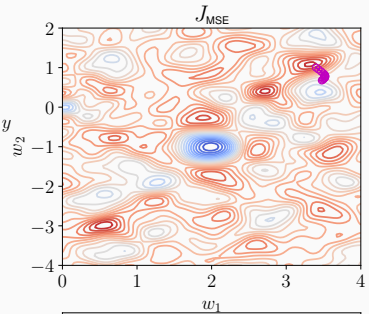
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.84$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

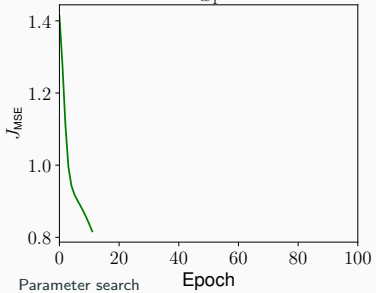
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

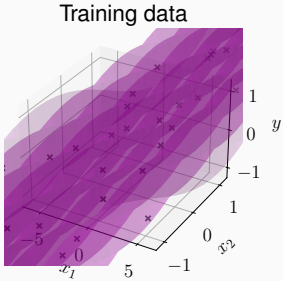
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

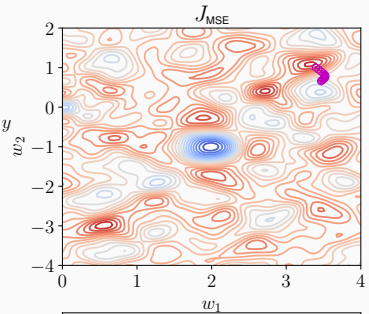
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.82$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

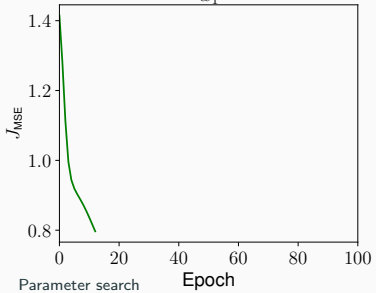
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

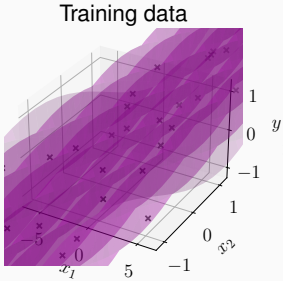
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

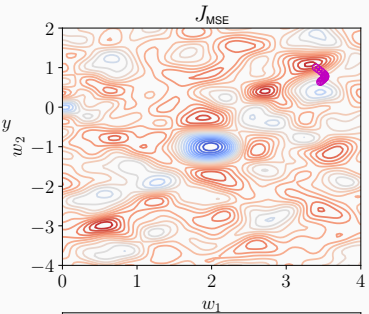
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.80$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

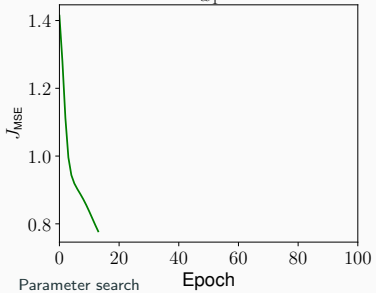
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

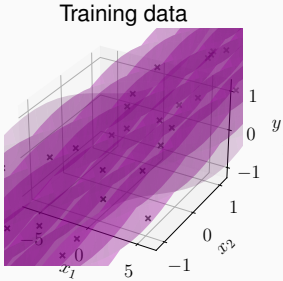
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

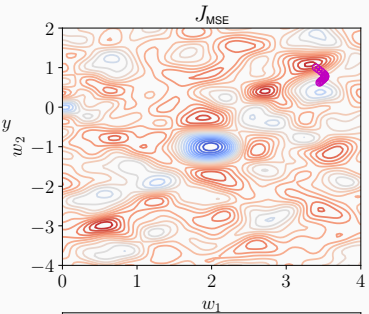
$$w_1 = 3.5, w_2 = 1$$

$$J_{\text{MSE}} = 0.78$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

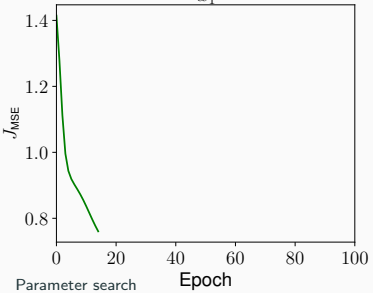
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

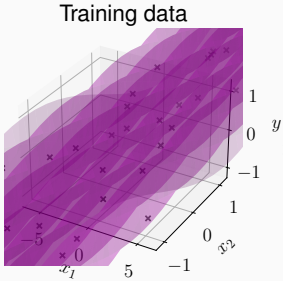
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

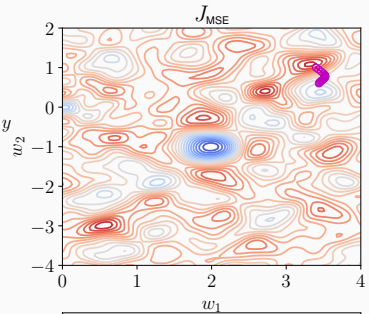
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.76$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

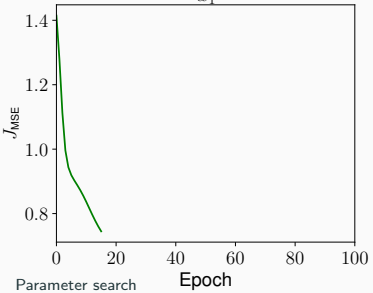
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

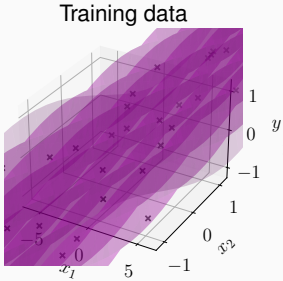
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

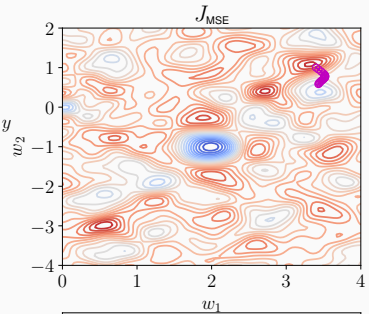
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.74$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

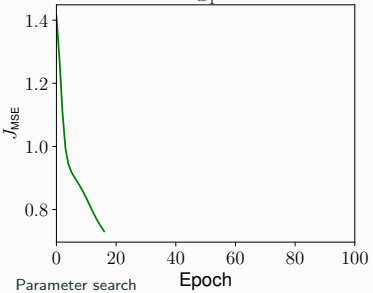
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

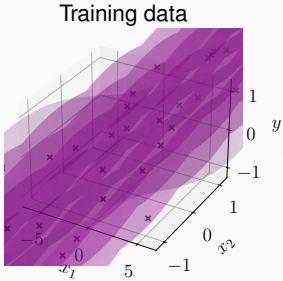
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

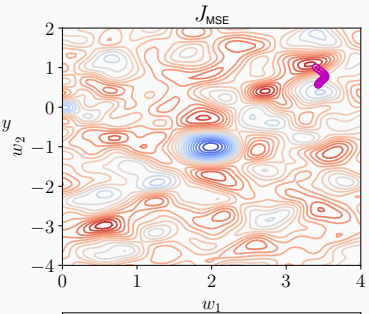
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.73$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

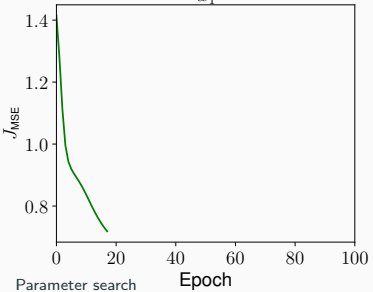
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

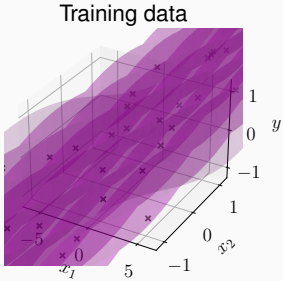
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

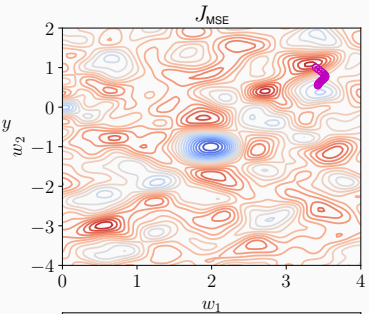
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.72$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

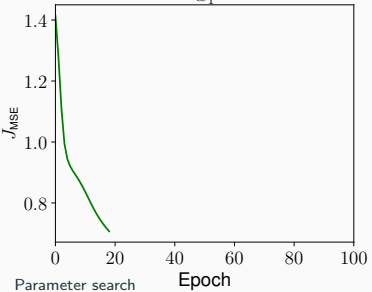
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

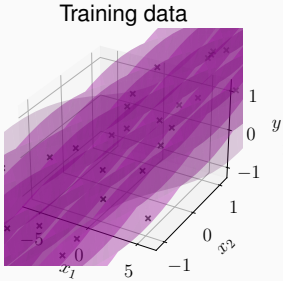
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

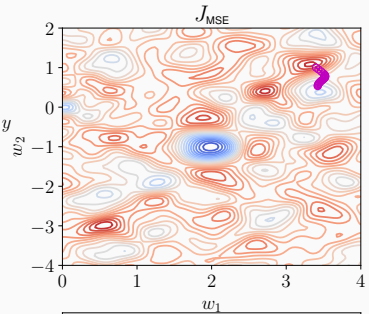
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.71$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

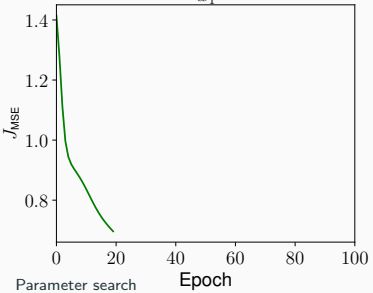
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

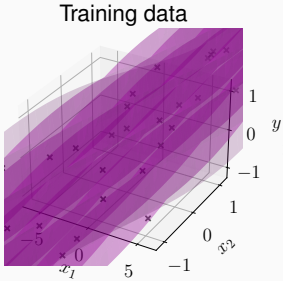
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

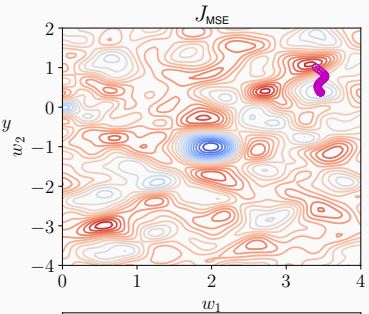
$$w_1 = 3.4, w_2 = 1$$

$$J_{\text{MSE}} = 0.70$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

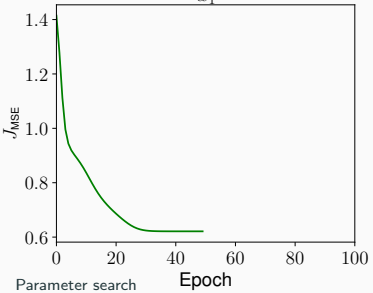
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

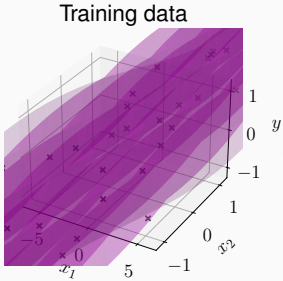
$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

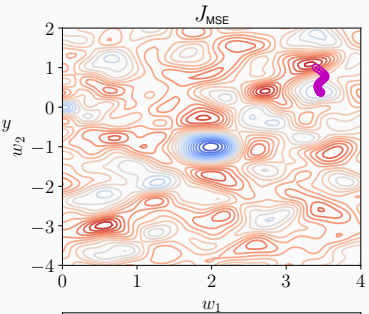
$$w_1 = 3.5, w_2 = 0$$

$$J_{\text{MSE}} = 0.62$$



Example: Steepest gradient descent non-linear regression ($\alpha = 0.1$)





$$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$$

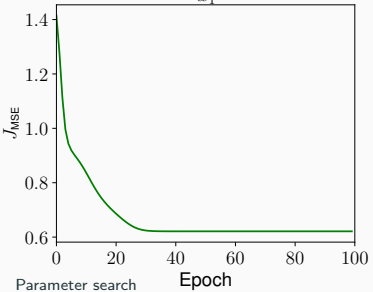
$$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$\Delta w_1 = -\frac{\partial J_{\text{MSE}}}{\partial w_1}$$

$$\Delta w_2 = -\frac{\partial J_{\text{MSE}}}{\partial w_2}$$

$$w_1 = 3.5, w_2 = 0$$

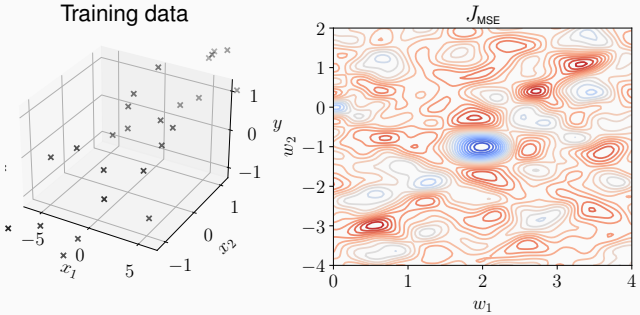
$$J_{\text{MSE}} = 0.62$$



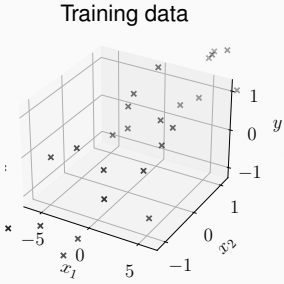
GA optimisation

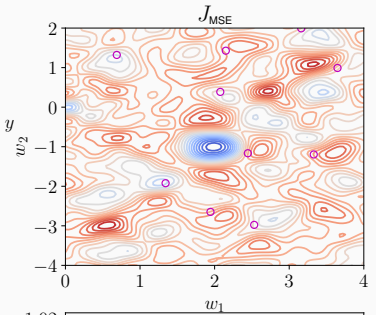
Stochastic search. We can create a **genetic algorithm** for searching the parameters of $h(x, w)$ that minimise some loss $J(w)$. For this setting, the set of different choices for w are random choice of a population of parameters, using the corresponding loss for each state and select the best state.

Example: GA optimisation non-linear regression (mutation rate = 0)



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

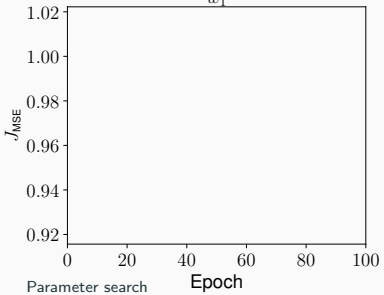
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

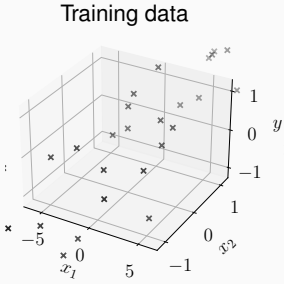
Elitism: 2, Mutation: 0.0

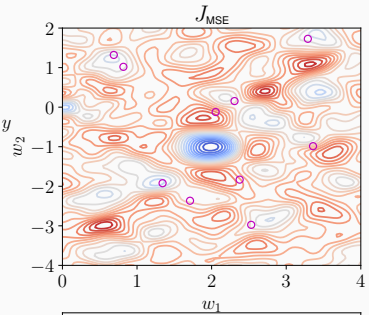
Best $w_1 = 3.2, w_2 = 2$

Best $J_{\text{MSE}} = 0.63$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

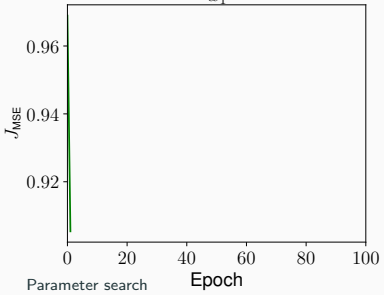
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

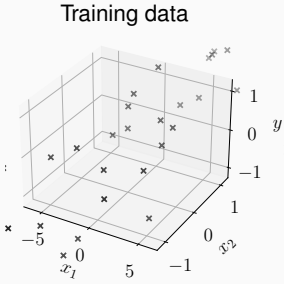
Elitism: 2, Mutation: 0.0

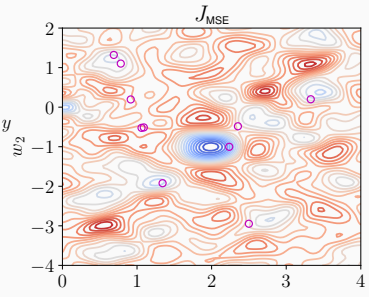
Best $w_1 = 1.3, w_2 = -2$

Best $J_{\text{MSE}} = 0.63$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

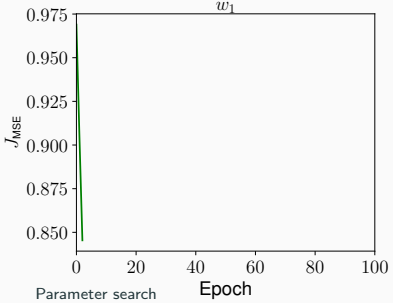
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

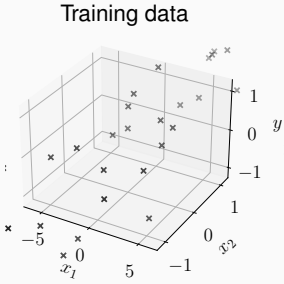
Elitism: 2, Mutation: 0.0

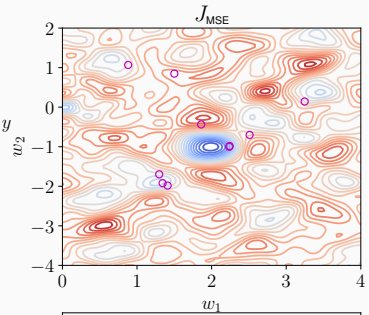
Best $w_1 = 1.3, w_2 = -2$

Best $J_{\text{MSE}} = 0.37$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

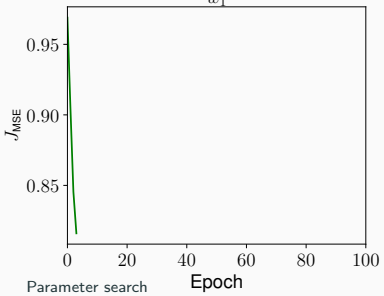
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

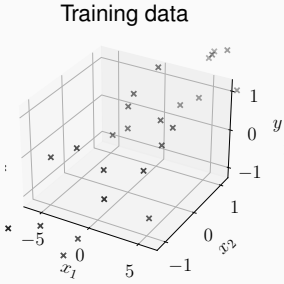
Elitism: 2, Mutation: 0.0

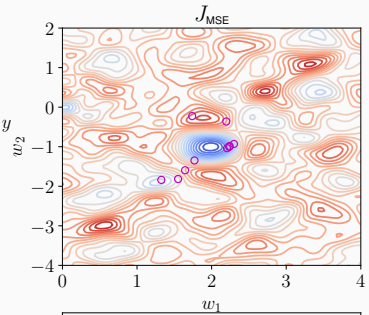
Best $w_1 = 2.2, w_2 = -1$

Best $J_{\text{MSE}} = 0.37$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

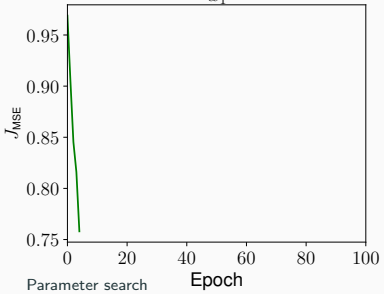
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

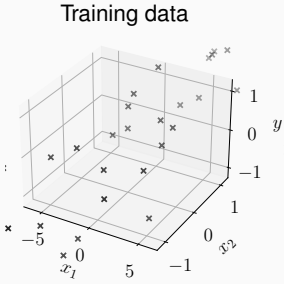
Elitism: 2, Mutation: 0.0

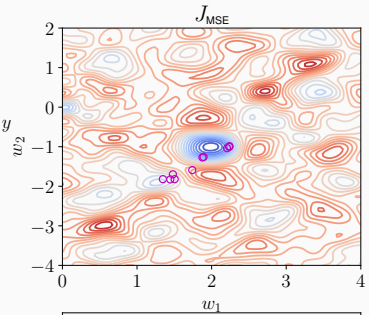
Best $w_1 = 2.2, w_2 = -1$

Best $J_{\text{MSE}} = 0.30$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

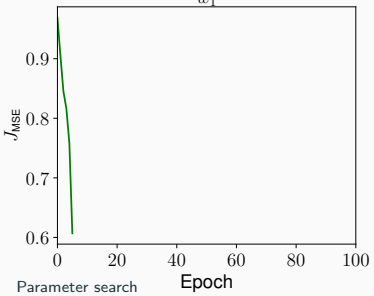
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

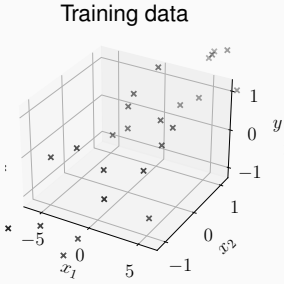
Elitism: 2, Mutation: 0.0

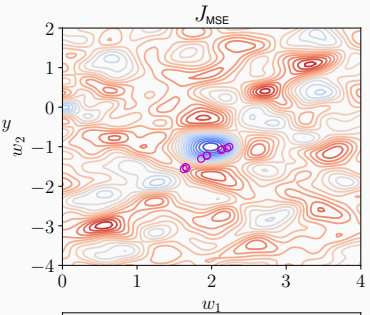
Best $w_1 = 2.2, w_2 = -1$

Best $J_{\text{MSE}} = 0.30$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

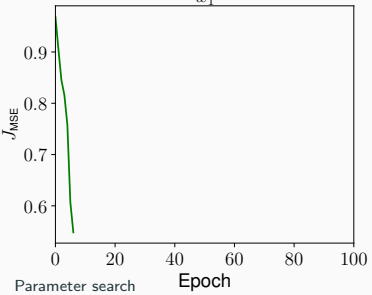
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

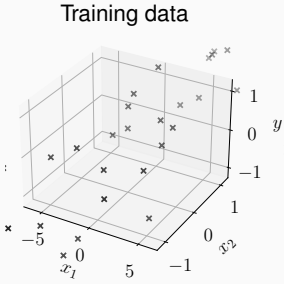
Elitism: 2, Mutation: 0.0

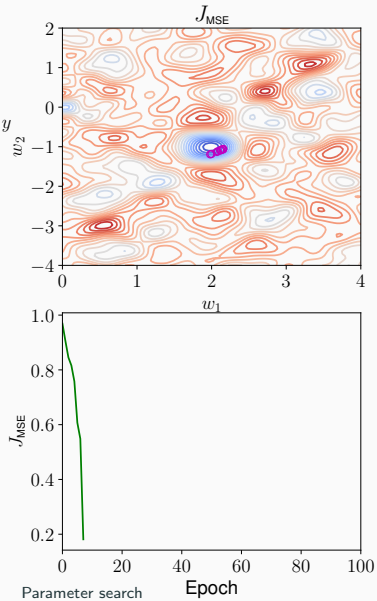
Best $w_1 = 2.2, w_2 = -1$

Best $J_{\text{MSE}} = 0.17$



Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

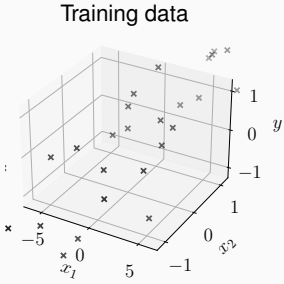
Population size: 10

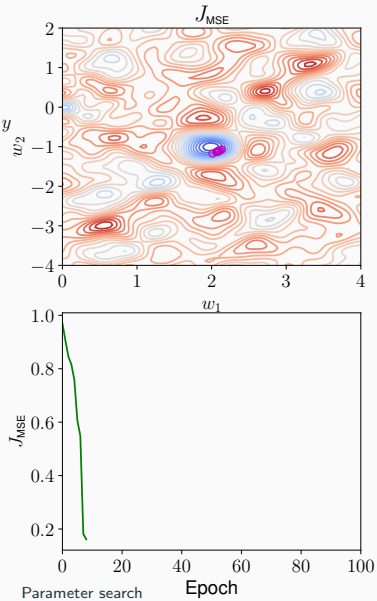
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

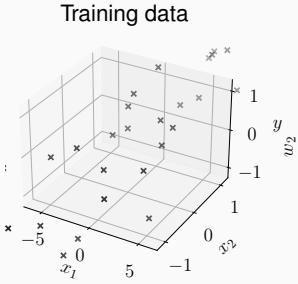
Population size: 10

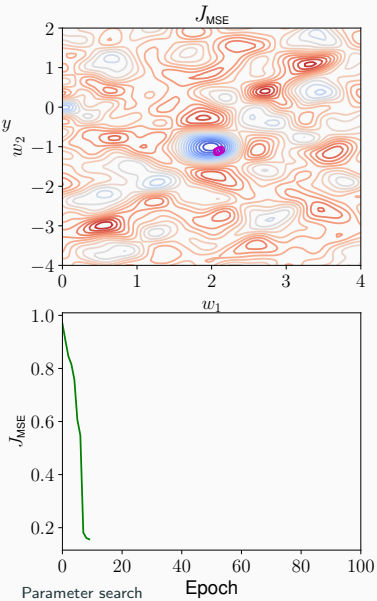
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

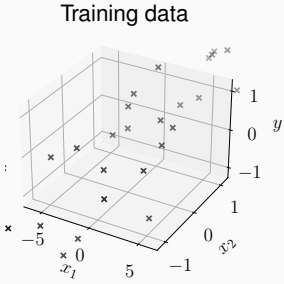
Population size: 10

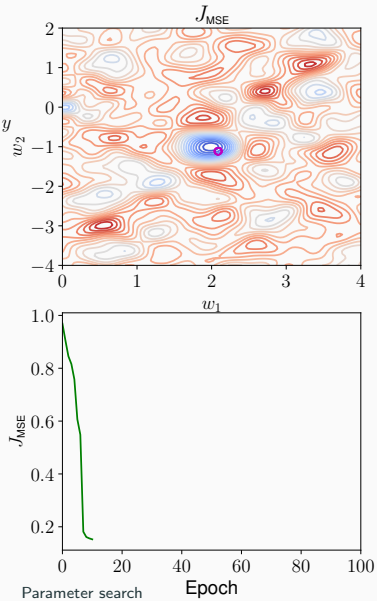
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

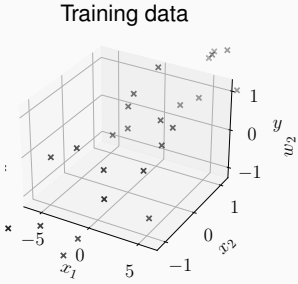
Population size: 10

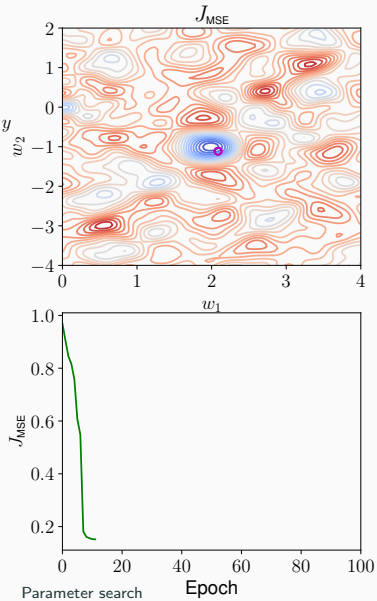
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

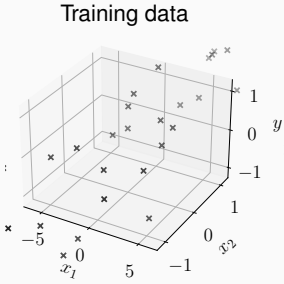
Population size: 10

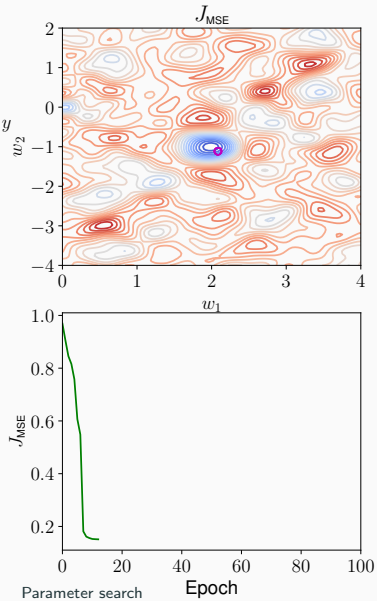
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

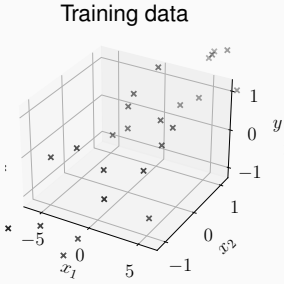
Population size: 10

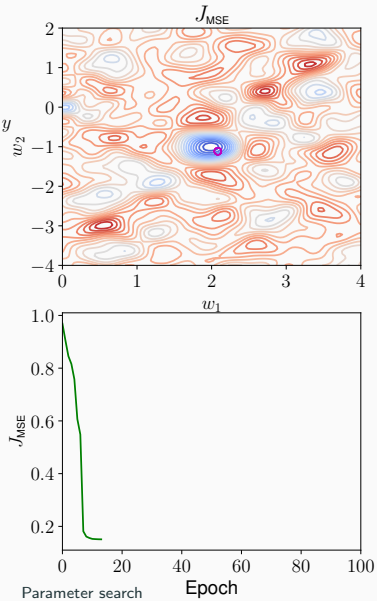
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y})^2$

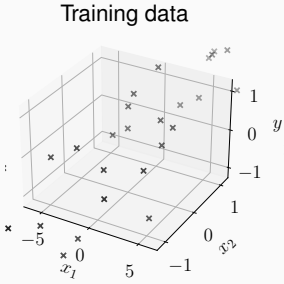
Population size: 10

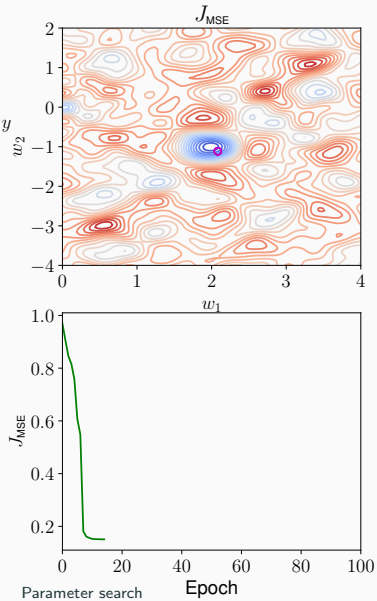
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

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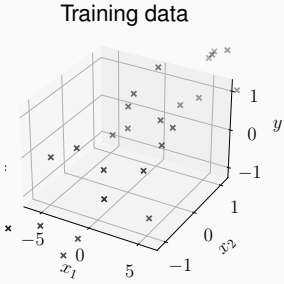
Population size: 10

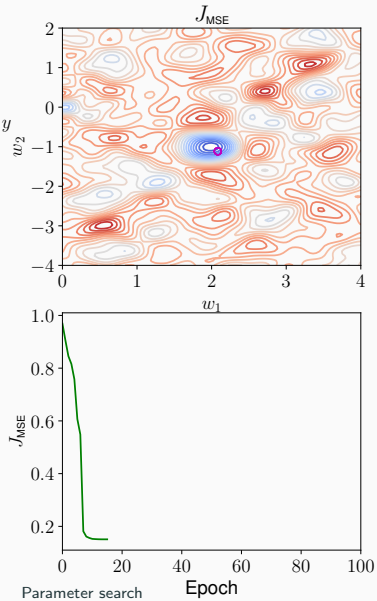
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

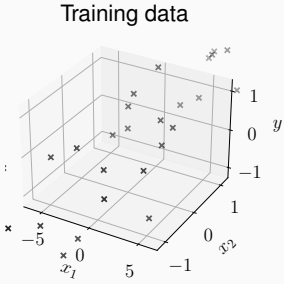
Population size: 10

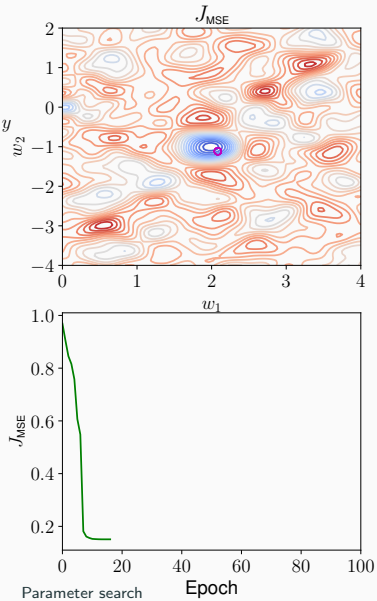
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

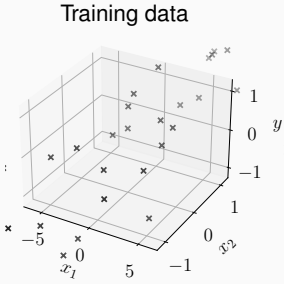
Population size: 10

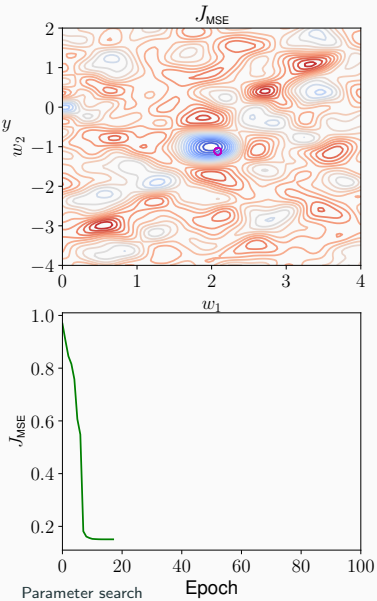
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

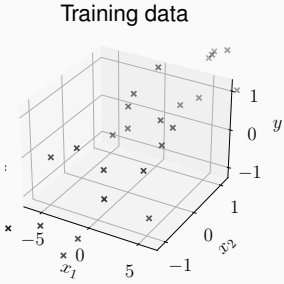
Population size: 10

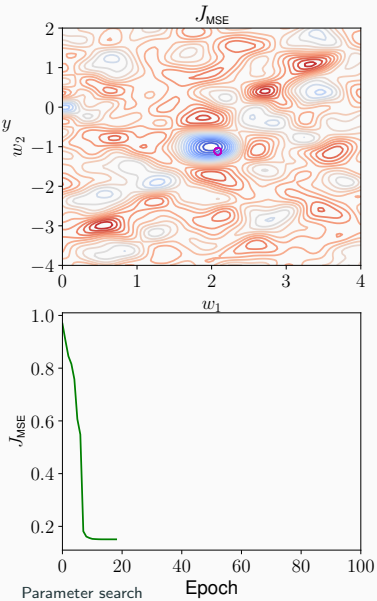
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

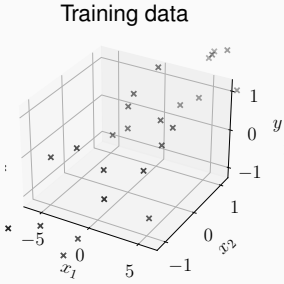
Population size: 10

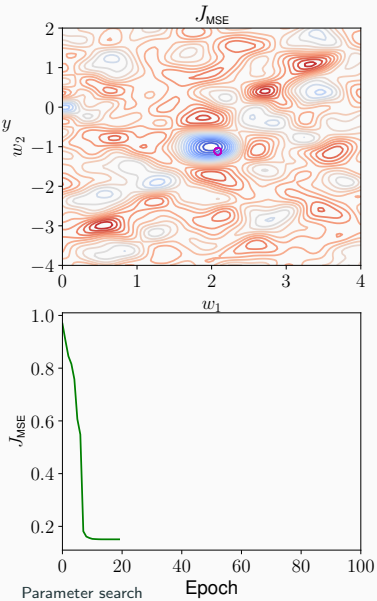
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

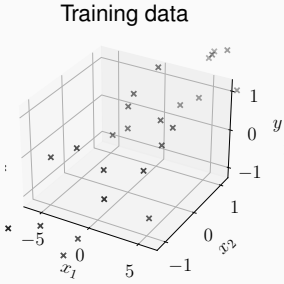
Population size: 10

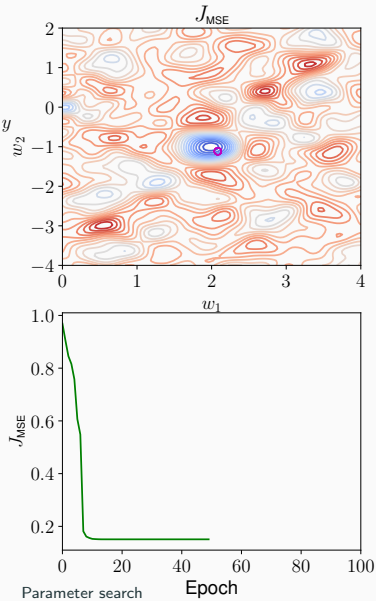
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

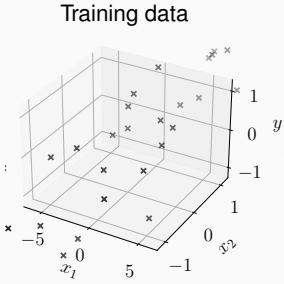
Population size: 10

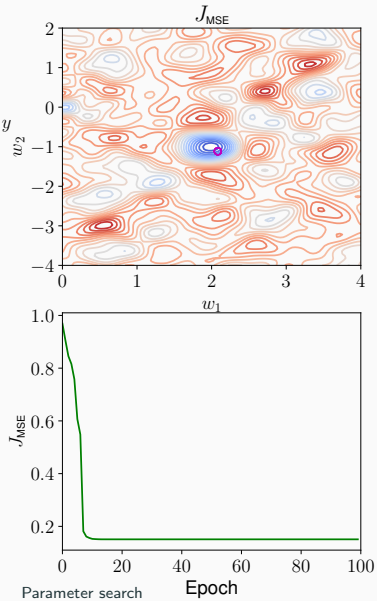
Elitism: 2, Mutation: 0.0

Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

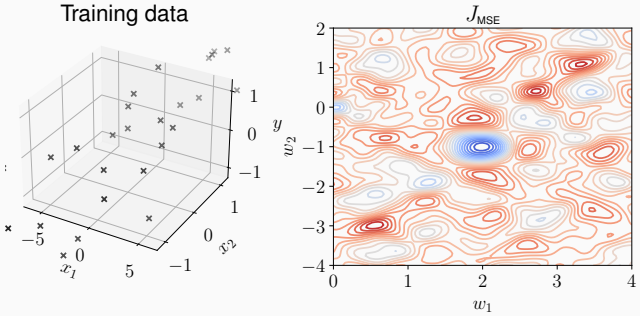
Population size: 10

Elitism: 2, Mutation: 0.0

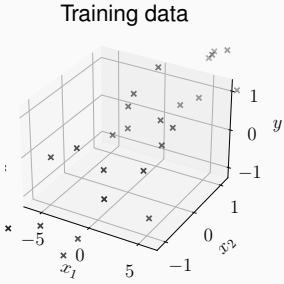
Best $w_1 = 2.1, w_2 = -1$

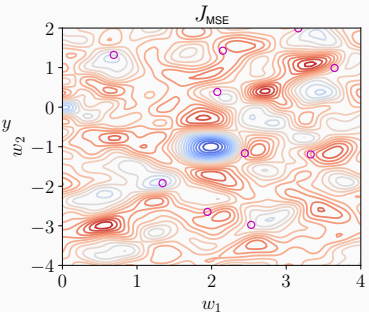
Best $J_{\text{MSE}} = 0.15$

Example: GA optimisation non-linear regression (mutation rate = 0.1)



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

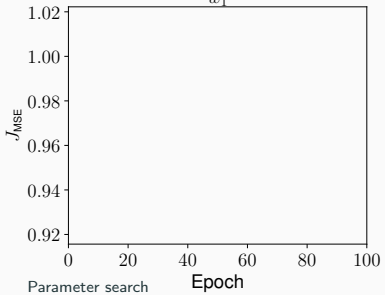
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

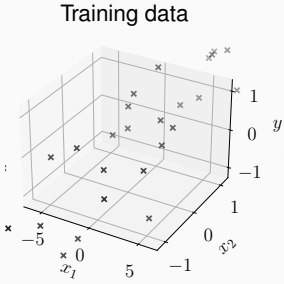
Elitism: 2, Mutation: 0.1

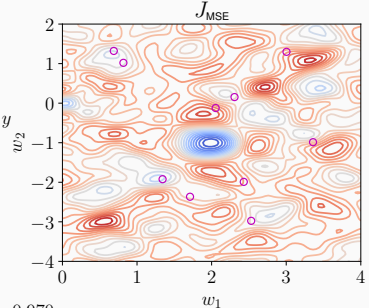
Best $w_1 = 3.2, w_2 = 2$

Best $J_{\text{MSE}} = 0.63$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

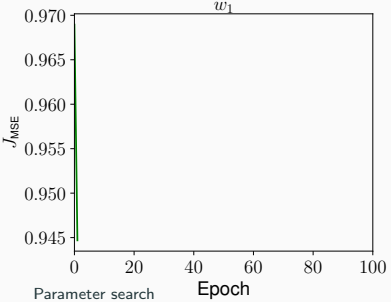
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

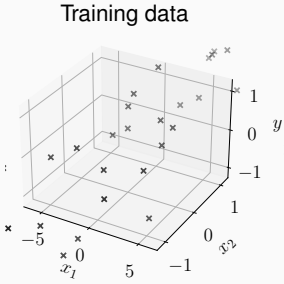
Elitism: 2, Mutation: 0.1

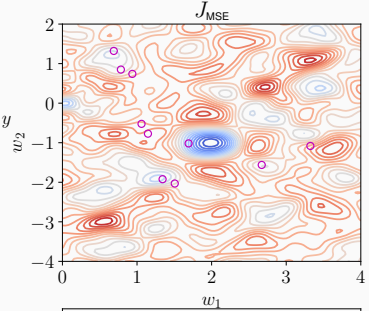
Best $w_1 = 1.3, w_2 = -2$

Best $J_{\text{MSE}} = 0.63$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

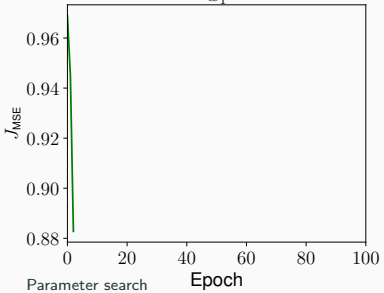
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

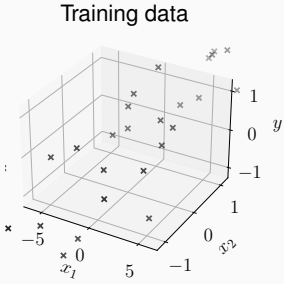
Elitism: 2, Mutation: 0.1

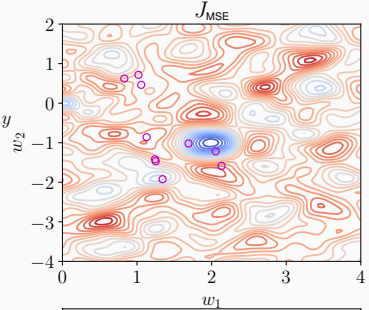
Best $w_1 = 1.3, w_2 = -2$

Best $J_{\text{MSE}} = 0.48$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

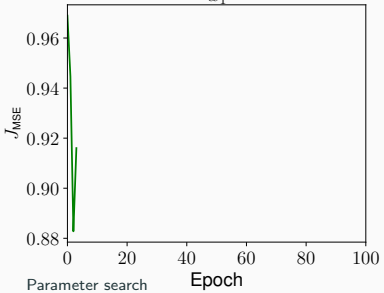
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

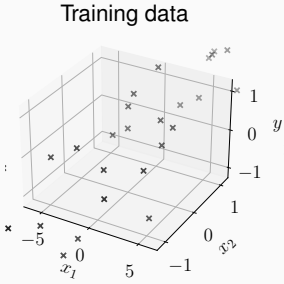
Elitism: 2, Mutation: 0.1

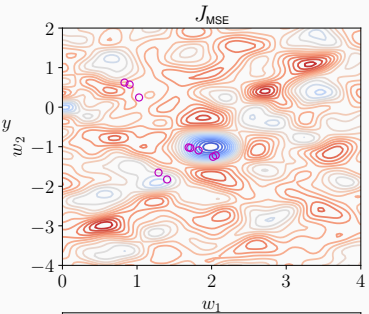
Best $w_1 = 1.7, w_2 = -1$

Best $J_{\text{MSE}} = 0.32$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

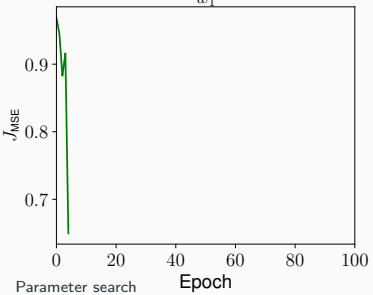
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

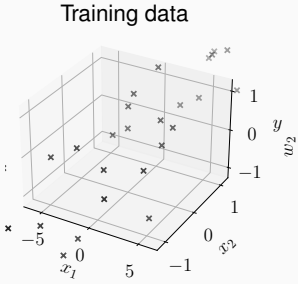
Elitism: 2, Mutation: 0.1

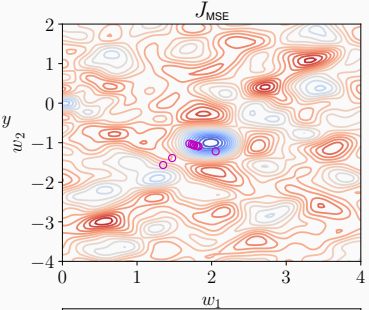
Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.19$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

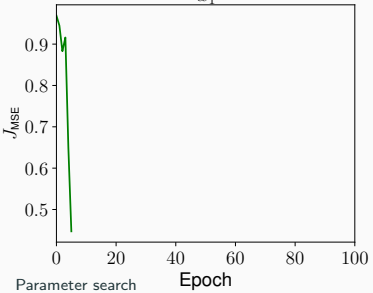
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

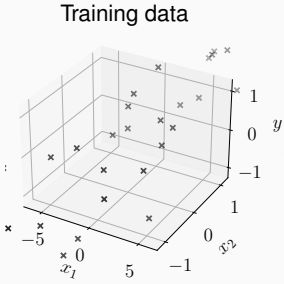
Elitism: 2, Mutation: 0.1

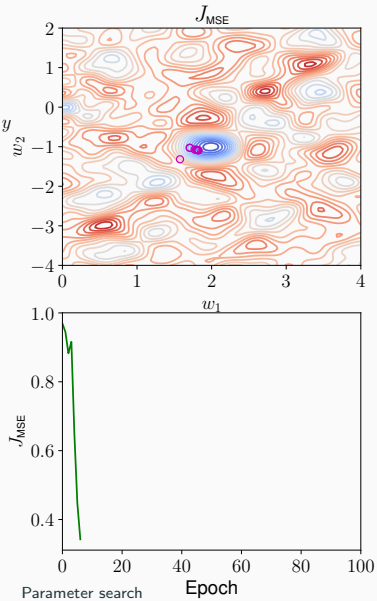
Best $w_1 = 1.8, w_2 = -1$

Best $J_{\text{MSE}} = 0.19$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

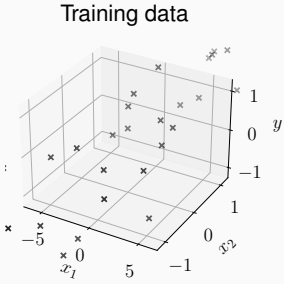
Population size: 10

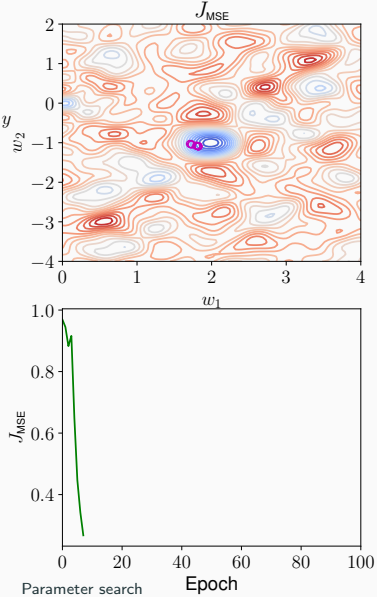
Elitism: 2, Mutation: 0.1

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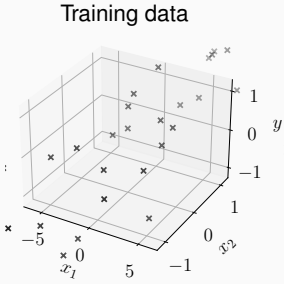
Population size: 10

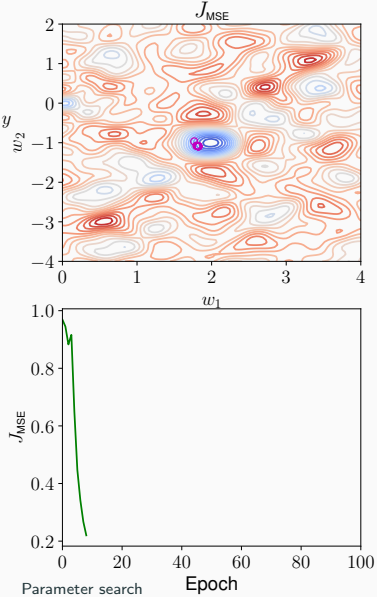
Elitism: 2, Mutation: 0.1

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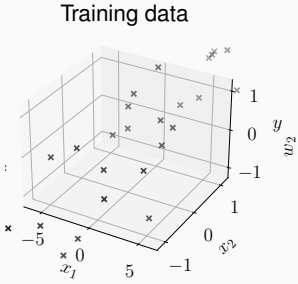
Population size: 10

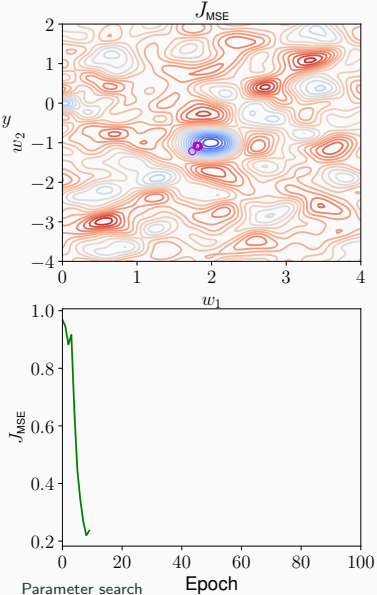
Elitism: 2, Mutation: 0.1

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Example: GA optimisation non-linear regression (mutation rate = 0.1)





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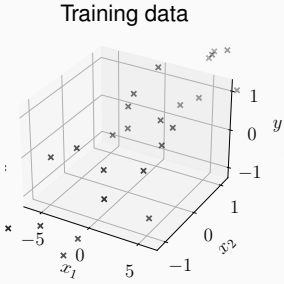
Population size: 10

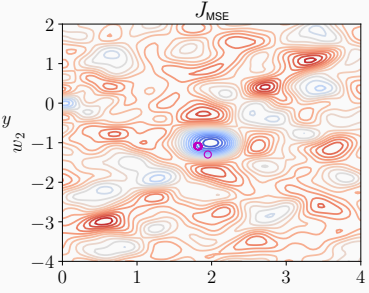
Elitism: 2, Mutation: 0.1

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Example: GA optimisation non-linear regression (mutation rate = 0.1)





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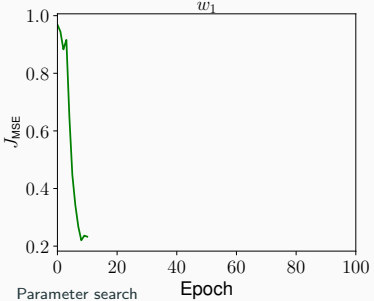
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Population size: 10

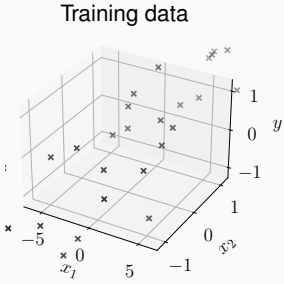
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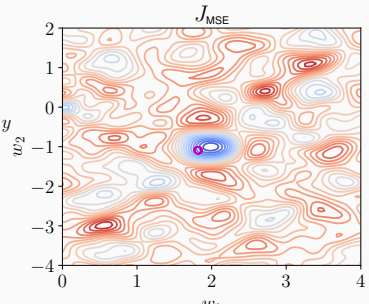
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Example: GA optimisation non-linear regression (mutation rate = 0.1)





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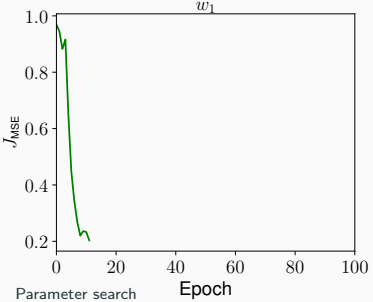
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Population size: 10

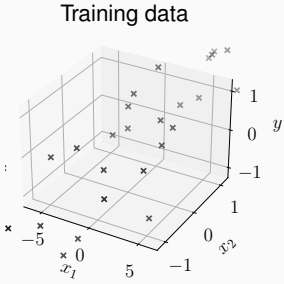
Elitism: 2, Mutation: 0.1

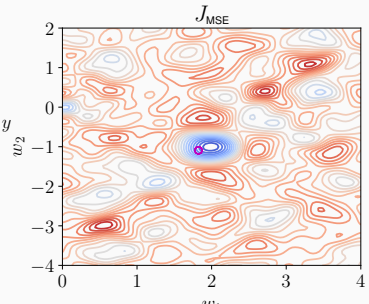
Best $w_1 = 1.8, w_2 = -1$

Best $J_{\text{MSE}} = 0.19$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

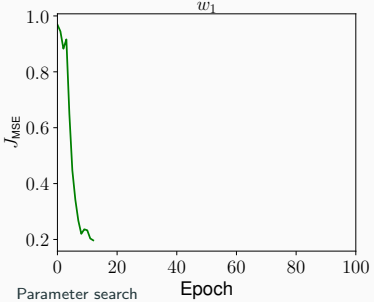
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

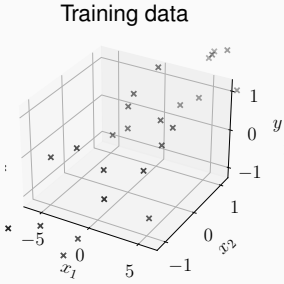
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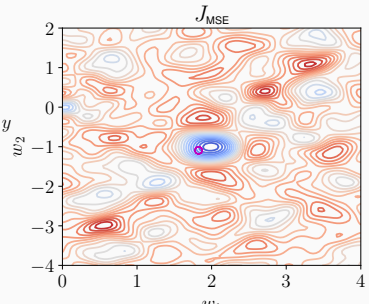
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Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

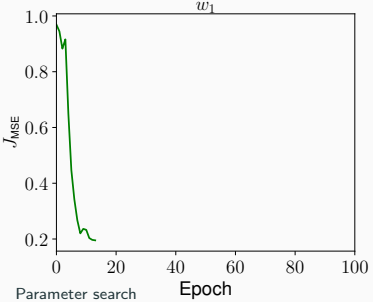
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

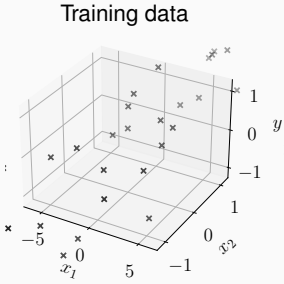
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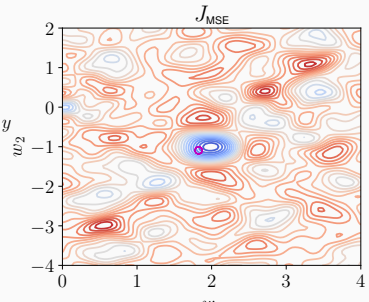
Best $w_1 = 1.8, w_2 = -1$

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Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

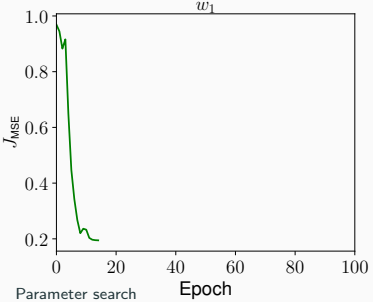
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

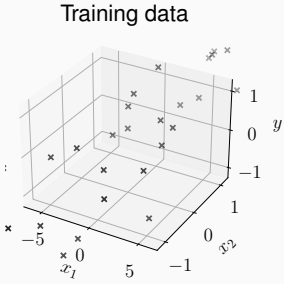
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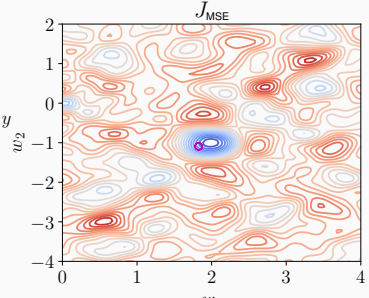
Best $w_1 = 1.8, w_2 = -1$

Best $J_{\text{MSE}} = 0.19$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

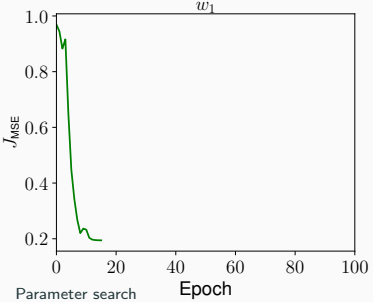
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

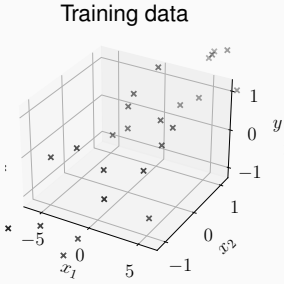
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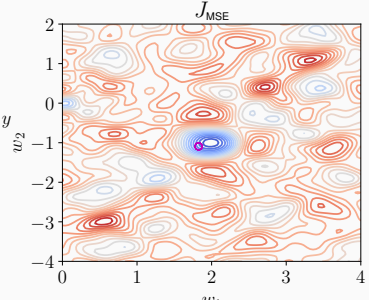
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Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

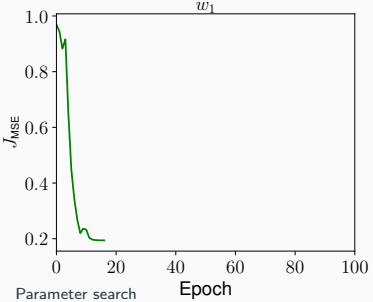
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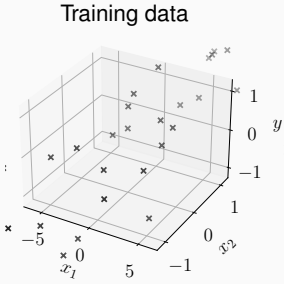
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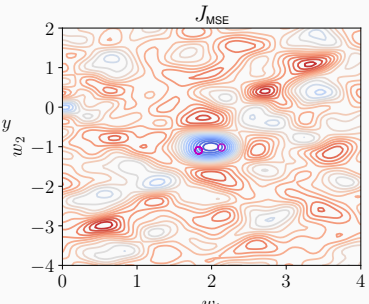
Best $w_1 = 1.8, w_2 = -1$

Best $J_{\text{MSE}} = 0.19$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

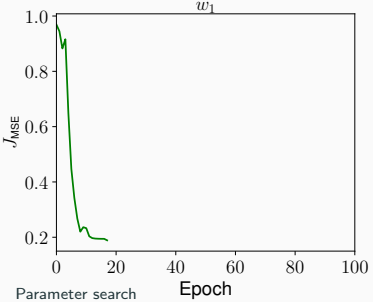
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

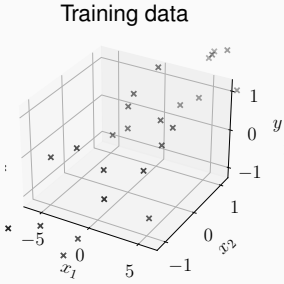
Elitism: 2, Mutation: 0.1

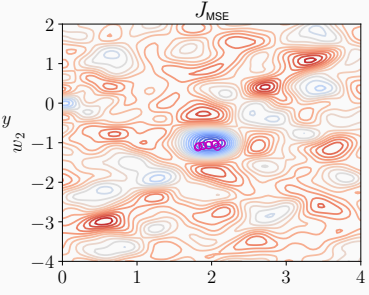
Best $w_1 = 1.8, w_2 = -1$

Best $J_{\text{MSE}} = 0.14$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

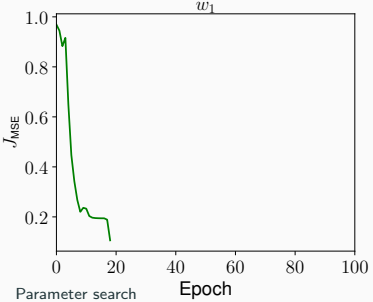
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Population size: 10

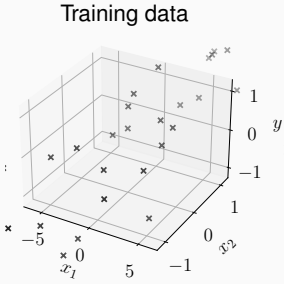
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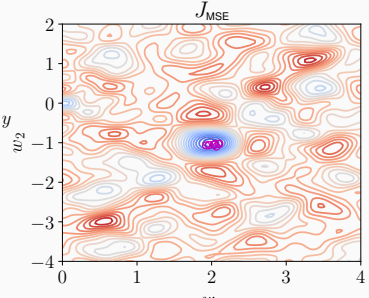
Best $w_1 = 2.1, w_2 = -1$

Best $J_{\text{MSE}} = 0.04$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

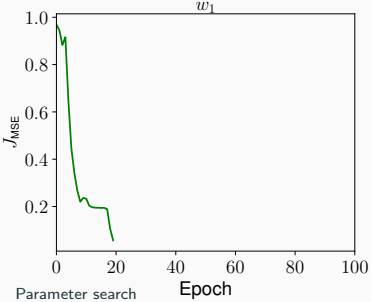
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

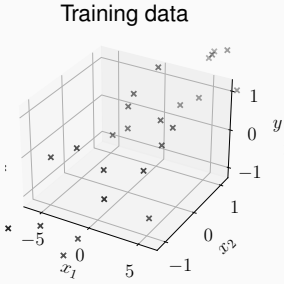
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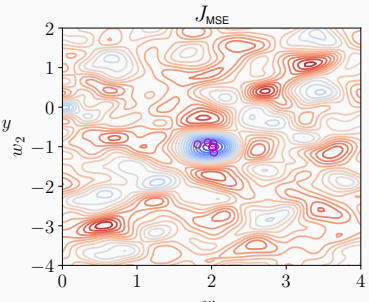
Best $w_1 = 2.0, w_2 = -1$

Best $J_{\text{MSE}} = 0.04$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

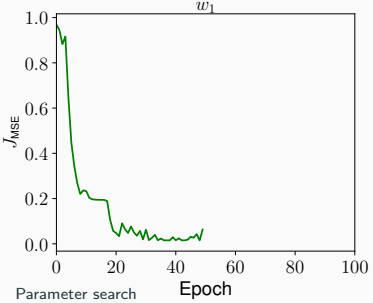
$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

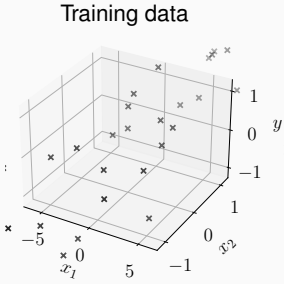
Elitism: 2, Mutation: 0.1

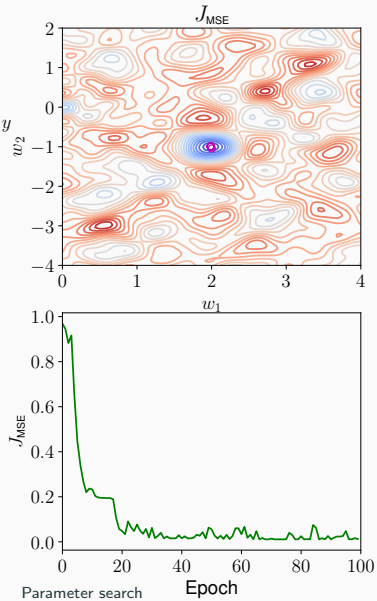
Best $w_1 = 2.0, w_2 = -1$

Best $J_{\text{MSE}} = 0.02$



Example: GA optimisation non-linear regression (mutation rate = 0.1)





$\hat{y} = \sin(w_1 x_1 + w_2 x_2)$

$J_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$

Population size: 10

Elitism: 2, Mutation: 0.1

Best $w_1 = 2.0, w_2 = -1$

Best $J_{\text{MSE}} = 0.01$

Study guide

- Understand the analogy of learning optimisation as parameter search
- Terms and definitions: loss surface, linear in parameters, convex loss, global and local minima, optimisation
- Understand how steepest gradient descent works (no need to know the math, but need to grasp what a gradient is, how it drives gradient descent, and what its limitations are)
- Understand how to use GA for learning parameters of a model.

Reading for this lecture

AIMA Ch.4.2,19.6.2

What's next?

Logistic regression

Read AIMA Ch.19.6.5